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TOPICAL REVIEW

A Survey of Industrial AIoT: Opportunities, Challenges, and Directions

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ABSTRACT Internet of Things (IoT) is an important technology employed in a variety of different applications, such as transportation, healthcare, and manufacturing. In recent years, the number of IoT devices deployed globally has been increasing at a rapid pace and is estimated to reach 20 billion by the end of 2025. In modern industry, IoT plays a pivotal role by monitoring the condition of industrial machines and, consequently, improving the efficiency of industrial processes. To optimize the efficiency of industrial IoT applications, various Artificial Intelligence (AI) techniques have been adopted, leading to a new computing paradigm, namely, Industrial Artificial Intelligence of Things (i.e. Industrial AIoT). In this paper, we describe the challenges to tackle and the opportunities to explore in Industrial AIoT. Specifically, we first review the use of state-of-the-art AI methods in Industrial AIoT applications, with a focus on Deep Learning (DL) and Machine Learning (ML) techniques. Thereafter, we present a series of important applications of Industrial AIoT. The key challenges associated with the implementation of Industrial AIoT applications are also discussed. In addition, the societal and economic impacts of Industrial AIoT are briefly described. Finally, we outline the future research directions in Industrial AIoT, which should be further investigated to fully utilize the potential of this innovative technology.

INDEX TERMS Internet of Things (IoT), industrial internet of things (IIoT), artificial intelligence (AI), artificial intelligence of things (AIoT), industrial AIoT, machine learning (ML), deep learning (DL).

I. INTRODUCTION

Internet of Things (IoT) is an important technology that can be integrated into a variety of different applications, such as environment monitoring, healthcare, transportation, agriculture, and manufacturing [1], [2], [3], [4], [5], [6], [7], [8], [9]. In the consumer market, one of the most well-known IoT applications is smart home [10], which involves intelligent devices, such as light switches and thermostats, that can be controlled via smartphones or smart speakers. In the healthcare domain, IoT has been applied to collect vital data and monitor patient health conditions, resulting in the Internet of Medical Things (IoMT) [11]. In the application of environmental monitoring [12], IoT has been used to monitor air/soil quality, and track the movement

of livestock and wild animals. An IoT system is composed of varied computing devices, often with built-in sensors that connect to the Internet or other types of networks for communication and data sharing purposes. Built-in sensors on IoT devices are used to collect data. Interconnection between IoT devices enables data exchange and remote control. Over the past years, the number of IoT devices deployed in the world has been increasing at a rapid pace. It is estimated that the total number of IoT devices will reach 20 billion by the end of 2025 [13].

It is worth noting that IoT is playing a pivotal role in modern industry [14]. With the deployed IoT devices, the performance and condition of industrial machines can be effectively monitored. In addition, industrial processes become more autonomous and consequently less human input is required. Finally, the IoT devices can help to optimize industrial processes, ultimately minimizing the

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overall manufacturing cost. In fact, the deployment of IoT in the industrial sector has led to a new computing paradigm known as Industrial Internet of Things (IIoT) [15]. Formally, IIoT is “the network of intelligent and highly connected industrial components that are deployed to achieve high production rate with reduced operational costs through real-time monitoring, efficient management and controlling of industrial processes, assets and operational time” [16]. The fourth industrial revolution, which is termed as Industry 4.0, particularly focuses on the automation of the manufacturing industry [16]. IIoT is one of the key enabling technologies for Industry 4.0 [17].

Artificial Intelligence of Things (AIoT) is another widely discussed computing paradigm that combines two cutting-edge technologies, Artificial Intelligence (AI) and IoT [18]. As mentioned previously, an IoT system typically consists of many devices with built-in sensors, which can generate an enormous amount of data. To efficiently process the collected data and thereafter make appropriate decisions, an effective data processing and decision-making mechanism is required. When AI is used to solve the data processing and decision-making problems in IoT, the resulting technology is called AIoT. Recently, there have been a number of attempts to apply AIoT to varied industrial domains to improve their efficiency [19], [20], [21], [22], [23]. Specifically, Lee et al. [21] found that AI can be used to help collect environmental data from disaster sites and coordinate robots to provide prompt disaster responses. Daoud and Otair [22] concluded that applying AI to IoT enables a new technology that can read medical images and provide error-free diagnostic results.

The application of AIoT in the industrial domain has exposed many opportunities to explore and challenges to tackle [24], which results in a novel technology, namely, Industrial AIoT. Fig. 1 shows the architecture of Industrial AIoT. The proposed architecture is composed of an industrial site, AI-enabled sensors, an edge gateway, and a control center. The industrial site is a geographical area that is specially designed for factories and manufacturing plants. It contains a variety of equipment, including robots, IoT devices, and manufacturing machines that generate data. AI-enabled sensors include all types of sensors that collect data and transmit it to the edge gateway. The AI-enabled edge gateway, a middleware layer between sensors and the control center, is mainly responsible for receiving, pre-processing, and forwarding data to the AI-enabled control center. Data pre-processing activities at the edge gateway involve data cleaning, reduction, and integration from multiple heterogeneous sensors. The AI-enabled control center receives data from the edge gateway, performs data processing to generate valuable information, and makes predictions using AI techniques. Technically, Industrial AIoT lies at the intersection of IIoT and AIoT. Fig. 2 depicts the relationship between IoT, IIoT, AIoT, and Industrial AIoT using a Venn diagram. For clarity, the key notations used in this paper are summarized in Table 1.

TABLE 1. Key Notations, their definitions, and descriptions.

Acronyms	Definitions	Descriptions
IoT	Internet of Things	The network of connected physical objects.
AI	Artificial Intelligence	A branch of computer science that enables computers and machines to mimic the human intelligence.
ML	Machine Learning	A sub-field in AI, which typically constructs a model to make predictions or decisions.
DL	Deep Learning	A Family of ML methods based on artificial neural networks which involve multiple processing layers.
IIoT	Industrial Internet of Things	The network of connected industrial components.
IoMT	Internet of Medical Things	The network of connected medical devices in healthcare to collect the patient data and monitor health.
AIoT	Artificial Intelligence of Things	The combination of AI and IoT to analyze IoT device's data and make intelligent decision.
Industrial AIoT	Industrial Artificial Intelligence of Things	The application of AIoT in the industrial domain.
PdM	Predictive Maintenance	PdM is a technique that continuously monitors the equipment's health and performance and detects abnormalities before the failure occurs.
ICS	Industrial Control System	An ICS is an electronic system used for controlling industrial processes.
CPS	Cyber-Physical System	A CPS manages interconnected systems between physical components and computation capabilities.
M2M	Machine-to-Machine	M2M refers to direct communication between devices through any communication channel, including wired and wireless.
IDS	Intrusion Detection System	An IDS is the software or hardware system that detect malicious activity or policy violations on a network or systems.
CMS	Condition Monitoring System	CMS monitors the condition of industrial components and is used for PdM.
FL	Federated Learning	FL is a ML technique that is used for training an algorithm across multiple decentralized devices (either edge devices or servers) keeping local data samples.
MEC	Mobile Edge Computing	MEC is a network architecture that extends the cloud-like services at the edge of the network.

Machine learning (ML) is a sub-field of AI that is widely applied in Industrial AIoT. Typically, ML constructs a model to make predictions or decisions [25]. Deep learning (DL) refers to a family of ML methods based on artificial neural networks (ANNs) [26]. Note that the networks used in DL are considered deep in the sense that they include multiple layers. Strictly speaking, DL is a component of ML. To emphasize the importance of DL in Industrial AIoT, in the rest of this paper, we use DL to refer to deep learning methods and employ ML to denote non-deep learning machine learning approaches. In practice, both DL and ML have been widely applied to a variety of different Industrial

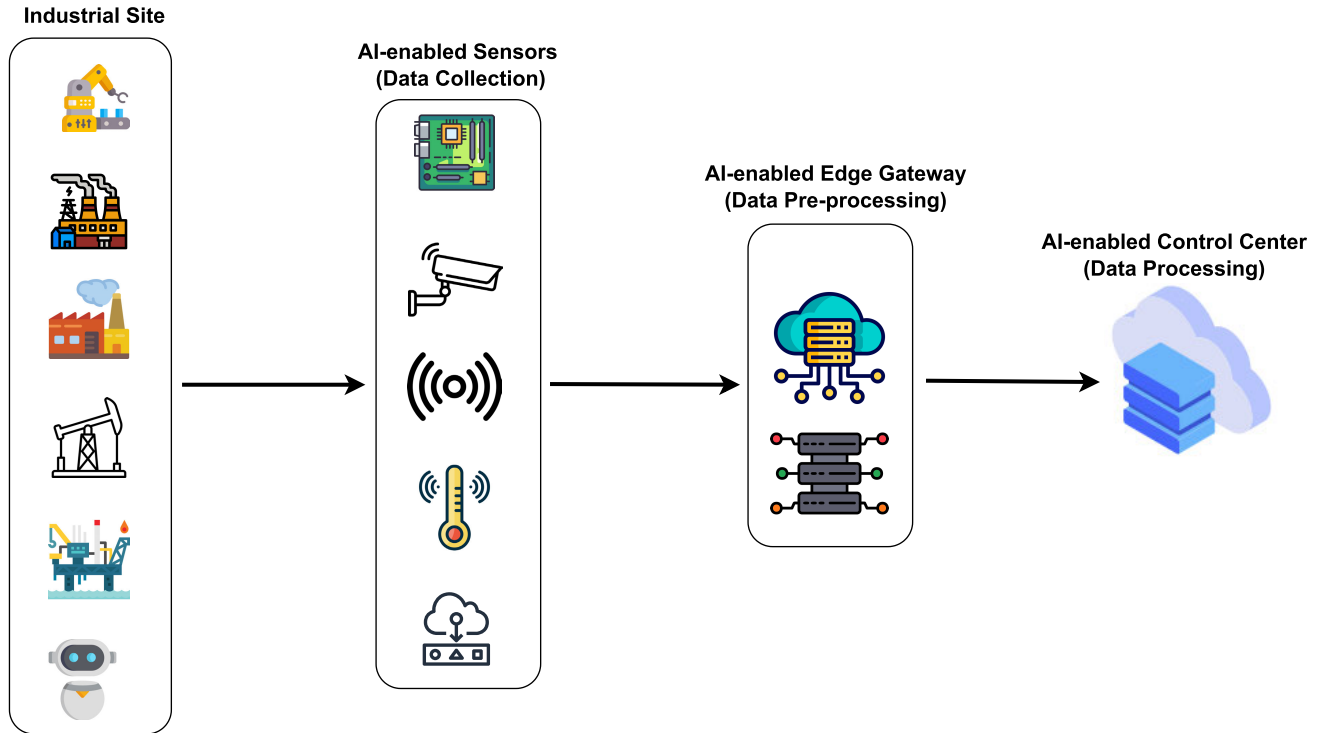


FIGURE 1. Architecture of industrial AIoT.

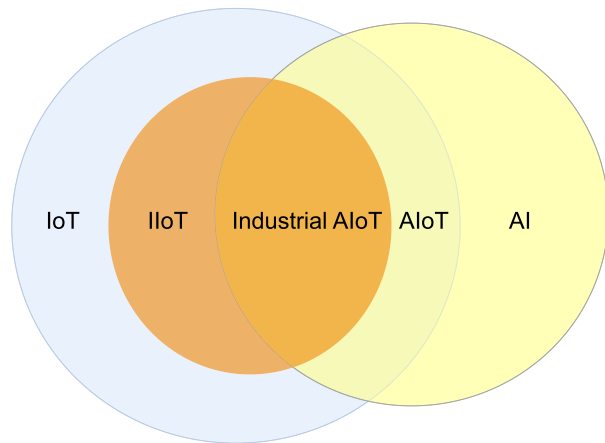


FIGURE 2. Relationship between IoT, IIoT, AIoT, and industrial AIoT.

AIoT tasks, such as making machine maintenance schedule, optimizing real-time manufacturing, and reducing energy consumption and downtime [27], [28], [29], [30]. In this paper, we will review the use of state-of-the-art AI methods in Industrial AIoT applications, focusing on DL and ML techniques. Specifically, we will cover ML techniques such as K-Nearest Neighbor (kNN), Support Vector Machines (SVM), Naive Bayes (NB), Decision Tree (DT), Random Forest (RF), Artificial Neural Network (ANN), and Federated Learning (FL). For DL methods, our discussion will include Autoencoder (AE), Convolutional Neural Network (CNN),

Recurrent Neural Network (RNN), Restricted Boltzmann Machine (RBM), Generative Adversarial Network (GAN), Deep Reinforcement Learning (DRL), Transformer, and Large Language Models (LLMs). Furthermore, we will identify opportunities and challenges associated with Industrial AIoT tasks. We will also provide a few research directions worth exploring in the upcoming years.

The structure of this paper is summarized in Fig. 3. The rest of this paper is organized as follows: Section II includes the related work on Industrial AIoT. In Section III, we review the fundamental DL and ML techniques in Industrial AIoT. The key applications of Industrial AIoT are presented in Section IV. In Section V, we describe the challenges associated with Industrial AIoT. Future research directions for Industrial AIoT are discussed in Section VI. Section VII highlights the societal and economic impact of Industrial AIoT. Finally, Section VIII concludes the paper.

II. RELATED WORK

This paper provides a detailed description of DL and ML techniques for Industrial AIoT, along with key applications, research challenges, and future directions. Several recent survey papers have been published in this area. We compare our paper to the recently published survey/review papers from six perspectives. Table 2 includes the specific comparison results. A summary of each survey/review paper involved in our comparison is presented as follows.

Bertolini et al. [30] made a significant contribution in the field by presenting the use of ML techniques in

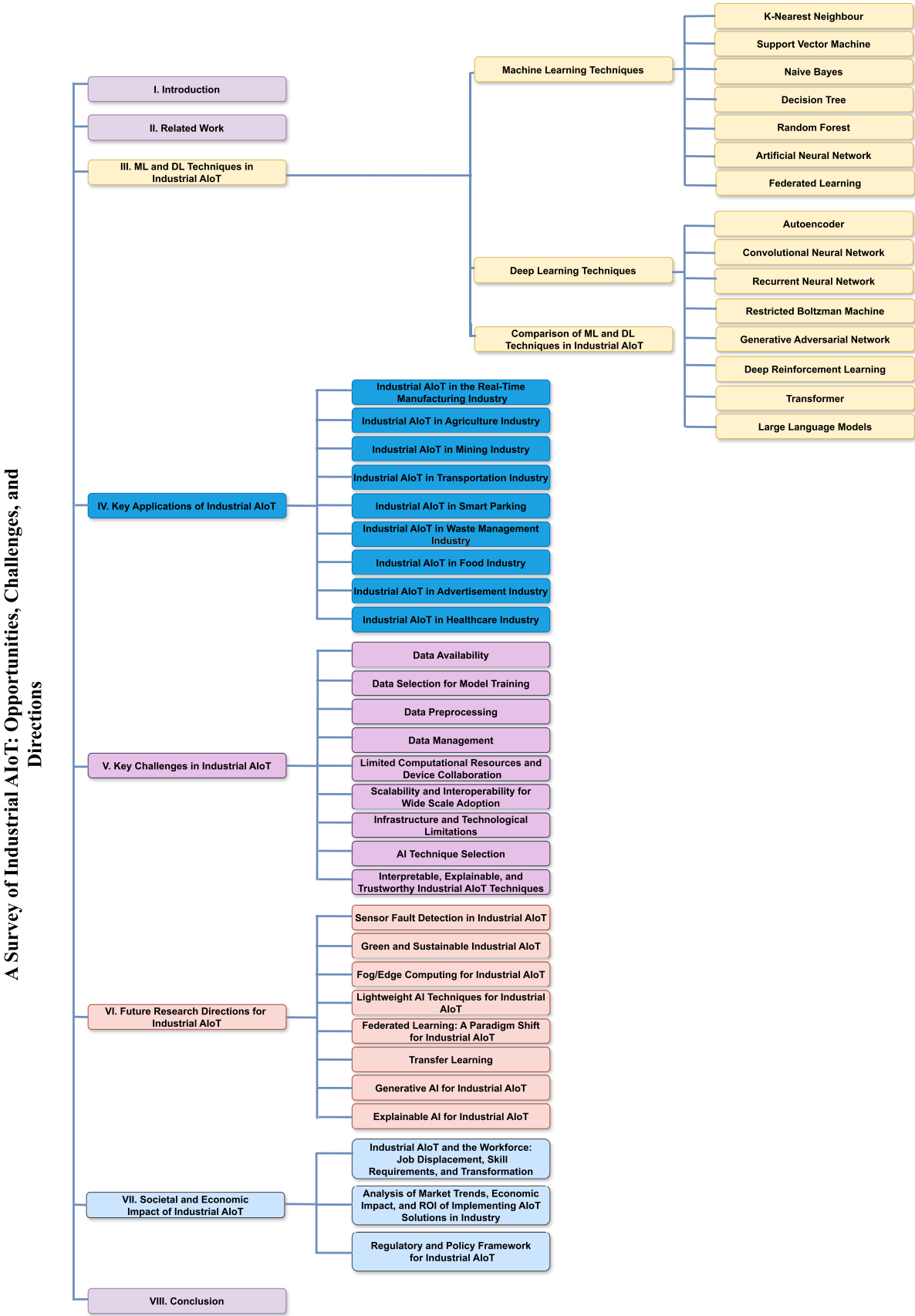


FIGURE 3. Structure of the paper.

industrial applications. The authors discussed ML core areas (supervised learning, unsupervised learning, reinforcement learning) in detail along with the systematic review methodology. They also highlighted the potential of ML in industrial operation management and described the prominent flaws. However, the scope of the paper is limited to ML techniques in operation management, such as supply chain management, production planning, and quality control. Furthermore, it does not discuss predictive maintenance (PdM) and other related operations in the industrial environment, nor does it cover all AI techniques, including DL.

Khalil et al. [24] discussed the importance and potential of DL in IIoT. First, the authors reviewed the various DL techniques and their use in industrial applications. They also presented the applications and challenges of DL in IIoT along with future research directions. However, the scope of the paper is only limited to the DL techniques in IIoT, and it does not discuss the ML techniques.

Latif et al. [31] emphasized the importance of IIoT and described the reference architecture. The authors discussed the popular DL techniques, their use in IIoT, and the various DL frameworks, including TensorFlow, Keras, Torch, PyTorch, etc. The authors also discussed the DL applications in IIoT and highlighted the challenges and future research directions. However, the scope of the paper is only limited to the DL techniques in IIoT and does not cover the ML techniques.

Silva et al. [32] reviewed the use of AI in five prominent industrial domains, including energy, transport sector, CPS, factories, and industrial processes. They presented a reference framework and architecture for research innovation and the use of AI in industrial informatics. Furthermore, they discussed a case study of AI in renewable energy and future directions of AI in industrial informatics. However, the study covers only five industrial sectors and does not discuss individual AI techniques in the overall industry in detail.

Peres et al. [33] presented a study on the use of AI in the industrial sector, primarily focusing on the manufacturing industry. They identified key AI technologies and design principles in the industrial sector, and discussed AI challenges, opportunities, and future research directions. The paper qualifies as one of the top studies describing Industrial AI techniques and identifying the ongoing research trends. However, it lacks a structured discussion of individual ML and DL techniques for IIoT.

Liu et al. [34] reviewed AI techniques for fault identification in rotating machinery, including kNN, NB, SVM, ANN, CNN, RBM, etc. They presented challenges and future trends of AI techniques for fault diagnosis in rotating machinery. However, the scope of the paper is limited to AI techniques for fault diagnosis in rotating machinery.

Ma et al. [35] conducted a survey of DL techniques for IoT applications. They surveyed the integration of DL and IoT domains to provide a novel design for IoT applications, discussing the use of DL techniques in various IoT domains like smart healthcare, home, transportation, and

industry. They identified challenges of implementing DL in IoT applications and provided future research directions. However, the paper only covers DL techniques.

In this survey, we contribute significantly to the field of Industrial AIoT by providing a detailed review of AI techniques, summarizing existing work in Industrial AIoT, describing key applications, highlighting key challenges, and outlining future research directions. The main contributions of the paper can be summarized as follows:

- We provide a detailed and comprehensive review of Industrial AIoT techniques and summarize the existing work related to its applications. Unlike existing surveys that focus solely on either ML or DL techniques in the industrial sector, our paper focuses on AI techniques, including both ML and DL approaches.
- We propose the Industrial AIoT Architecture, an integrated framework that combines key components such as industrial sites, smart sensors, gateway devices, and control centers to improve both operational efficiency and decision-making processes.
- We present key applications of Industrial AIoT, including real-time manufacturing, agriculture, and transportation. These applications are widely discussed in existing research on Industrial AIoT.
- We highlight the key research challenges associated with Industrial AIoT, including data selection, limited computational resources, and AI technique selection.
- We suggest emerging research directions in accordance with various Industrial AIoT challenges. The proposed research directions need to be thoroughly investigated to successfully integrate AIoT into the industrial sector.

III. ML AND DL TECHNIQUES IN INDUSTRIAL AIOT

Industrial AIoT has revolutionized the industrial sector, particularly the manufacturing industry [36]. By integrating AI, encompassing both ML and DL techniques, Industrial AIoT brings intelligence to industrial processes and machines. It connects a wide range of industrial devices, including machines, business processes, collaborative robots, and information systems. The data generated from various manufacturing processes, devices, sensors, and machines are processed for intelligent decision-making. This processing includes activities such as data analysis, modeling, and labeling. AI techniques, particularly ML and DL, are crucial in this context. ML has been a key factor in transitioning the traditional manufacturing industry to Industry 4.0. Specifically, ML techniques can be classified into supervised, unsupervised, and semi-supervised learning, which mainly solves the clustering, regression, and classification problems [37]. In contrast, DL techniques, which involve more advanced and computationally intensive methods, excel in automatic feature learning. While ML techniques may not be ideal for processing large volumes of data, DL techniques can effectively handle and yield satisfactory results with such data [24]. Considering the importance of ML as a sub-field

TABLE 2. Comparison of our paper and related survey/review papers.

Main Contribution	Our Pa- per	[30]	[24]	[31]	[32]	[33]	[34]	[35]
Discusses ML techniques for industrial applications	✓	✓	N	N	✓	✓	✓	N
Discusses DL techniques for industrial applications	✓	N	✓	✓	✓	✓	✓	✓
Covers multiple industrial perspectives (e.g. PdM, manufacturing, and anomaly detection)	✓	N	✓	✓	N	N	N	N
Presents key applications and use cases	✓	✓	✓	✓	N	✓	✓	✓
Lists research challenges	✓	✓	✓	✓	✓	✓	✓	✓
Points out future directions	✓	✓	✓	✓	✓	✓	✓	✓

of AI, this paper also discusses various ML techniques. Tables 3 and 4 summarize the key contributions of ML and DL techniques in Industrial AIoT, respectively.

A. MACHINE LEARNING

In this section, we discuss the state-of-the art ML techniques adopted in Industrial AIoT.

1) K-NEAREST NEIGHBOR (KNN)

kNN is a well-known supervised learning algorithm widely used in various applications for solving both regression and classification problems. However, it is more popular for solving classification problems. kNN assumes the similarity between the existing and new data and places the similar data in the same categories. kNN is broadly used in Industrial AIoT applications for solving various problems [38], [39], [40].

kNN has been mainly used in Industrial AIoT applications for defect identification, PdM, and anomaly detection purposes [41]. Nevertheless, kNN requires high computational resources in case of an enormous amount of data with many features. Several researchers have used the kNN in Industrial AIoT to perform various tasks, as outlined [41], [42], [43].

Yang et al. [41] proposed a secure and efficient distributed kNN (SEED-kNN) algorithm for Industrial AIoT. The proposed SEED-kNN algorithm classifies the large-scale encrypted data based on the similarity score. It is highly suitable in industrial control systems (ICSs) for applications such as defect identification, fault, and anomaly detection.

Marino et al. [42] presented a hybrid Fisher Wrapper approach for feature selection. Furthermore, the authors have applied two supervised algorithms as classifiers, i.e., kNN and SVM. The experimental results demonstrate that both algorithms work well for fault and anomaly detection.

Derhab et al. [43] presented a blockchain-based security enabled architecture for Industrial AIoT. The proposed security architecture combined the Random Subspace Learning (RSL) and kNN algorithms for efficient intrusion detection. The experimental results demonstrate that the RSL-kNN classifier achieves better accuracy and outperforms the traditional ML approaches.

ICSs can reduce human efforts by minimizing the interaction between the human and the control systems. kNN has been used as an efficient anomaly detection algorithm in ICS and Cyber-Physical System (CPS). The term CPS refers to technologies that manage interconnected systems between physical components and computation capabilities [44]. Selim et al. [45] used ML for Malicious Activities Recognition in Water-based Industrial AIoT. The authors applied the Genetic-kNN-based algorithm to identify malicious activities and anomalies in the industrial environment.

Since the last decade, tremendous advancements have been made in the industrial sector, especially the advent of M2M infrastructure. Consequently, the safety, reliability, and efficiency have become prominent concerns in the CPS and industrial environment. Ghosh et al. [46] proposed a secured edge gateway microservices architecture (SEGA) for monitoring the machine data in Industrial AIoT. SEGA monitors the different parameters, including power consumption, vibrational state, and predicts the current state of the machines. The authors have used the kNN classifier on the edge gateway that processes the machine's data to identify the machine state. Gao et al. [47] presented the Euclidean distance-based-between-class learning algorithm for intrusion detection in CPS that calculates the distance between different classes of kNN. Similarly, a modified kNN technique is presented in the [48] for the data-driven failure analysis for the CPS infrastructure. Experimental results show that the modified kNN approach achieves 99% accuracy for failure detection.

Due to the advent of new wireless technologies like IEEE 802.15.4, and ZigBee, there has been a considerable advancement in the industrial sector in the last decade. Therefore, the extensive use of technology in the industrial sector has made the industrial infrastructure so complex that anomalies and machine failures have become common problems. Raposo et al. [49] evaluated the performance of different ML algorithms, including kNN, to detect security issues and anomalies in the Industry. Durazno et al. [50] presented the energy monitoring-based ML approach for anomaly detection in a clean water supply system using kNN. The results demonstrate that the kNN is one of the best

classifiers for detecting anomalies with high accuracy and low false positive cases.

2) SUPPORT VECTOR MACHINE (SVM)

SVM is a classical ML algorithm for solving classification and regression problems. The SVM algorithm finds the hyperplane in the N-dimensional space to classify the data points. For instance, to separate the two data points, we can choose various hyperplanes where the purpose is to obtain a plane with the maximum distance between data points of both classes.

The use of AI techniques in the agriculture domain can help improve agriculture products' production and quality. The agriculture sensors generate a vast amount of data, and it is necessary to mine the valuable information from this massive data. Ruan et al. [51] combined the SVM with the genetic algorithm (GA) to analyze the big data in agriculture. The GA-SVM classifier provided better prediction accuracy in the agriculture CPS.

Zhang et al. [52] presented an SVM-based intelligent approach for combustion fault diagnosis for V-type marine diesel engines. They proposed a cyclic-polar map (CPM) to display the faults. The features were extracted using CPM, and the SVM classifier automatically detected the combustion faults from the extracted features. Similarly, Zhou et al. [53] discussed the SVM-based health monitoring and fault detection system for the ship systems. The proposed SVM technique combined both data and model-based approaches.

Large CPSs like the power plants often deploy the open network for various purposes, i.e., monitoring and control. The open network deployment gives rise to the cyberattack risks like denial-of-service (DoS) and replays attacks. To solve these types of issues in large power plant's infrastructure, Kordestani et al. [54] proposed a security management system (SMS) for the power turbine-based industrial system. The proposed SMS consists of attack detection and isolation units. The SVM classifier is used in the attack isolation unit, which detects the type of attack from the sensor data. The experimental results demonstrate that the proposed SMS approach achieves higher accuracy as compared to state-of-the-art techniques. Similarly, Khan et al. [55] presented an intelligent multi-class SVM-based anomaly detection technique in CPS inverter-based systems.

The rapid advancement in technology has led to the magnificent combination of IIoT and Computer Vision (CV) that is now widely used for workers' safety in the industrial sector. Wang et al. [56] presented the vision-based technique for identifying workers in industrial environment. Industrial workers use helmets to protect them from head injuries. The different color helmets are assigned to the workers to identify the job nature and type of the worker. The proposed technique uses the SVM classifier to extract the region of interest images and intelligently identify the worker type based on

the color of the helmet. Likewise, in [57], a quantum SVM-based approach is proposed for identifying human activity in Industry 4.0.

Recently, peer-to-peer (P2P) networks have emerged in Industrial AIIoT for distributed computing and ICS. One of the prominent characteristics of the P2P network is that they are independent, which gives rise to many security issues, i.e., anonymous access. To solve these issues, Alkhalaf [58] presented an authentication mechanism for making the P2P industrial systems more secure. The proposed scheme used the SVM to identify the machine access time and perform the group-based classification. Similarly, Xie et al. [59] used the SVM classifier to perform the clustering in the P2P network.

3) NAIVE BAYES (NB)

NB is a popular probabilistic ML algorithm used for solving classification problems [60]. It is based on Bayes' theorem and calculates probabilities based on recurrence and combinations of values in the given dataset. NB assumes that all variables in the dataset are independent. In the medical field, NB finds extensive applications in the detection of diseases such as heart disease, thyroid disorders, and COVID-19 [61], [62], [63], [64], [65]. However, its use in Industrial AIIoT for fault and anomaly detection is limited

For instance, Bhatti et al. [66] presented an intelligent fault detection mechanism for robots and actuators in the industrial environment using digital twin technology. The proposed fault detection model utilizes the NB algorithm to categorize data into different classes and identify the presence of faults.

The ratio of threats in CPSs is increasing due to the use of modern computing devices. Conventional intrusion detection systems (IDSs) are often insufficient for detecting intrusions in modern cyber-physical systems. Kreimel et al. [67] introduced a modern anomaly-based approach to detect and classify intrusions in CPS and industrial systems. This approach initially defines a model for the expected behavior of the system, calculating outlier scores for normal operations. Subsequently, new data is compared to the model to detect anomalies.

In addition, an attack model employing a trained NB classifier detects anomalies within the trained classes. This attack model is trained using well-known cyberattacks such as Man-in-the-middle, Dictionary, Denial-of-Service, and communication protocol attacks. To assess the performance of the proposed system, a test environment with various controllers, sensors, and actuators is created, generating data similar to actual data. Jayalaxmi et al. [68] presented a combination of DL and ML techniques for feature extraction in Industrial AIIoT. The authors used the Gas Pipeline and Water Storage datasets [69] for feature extraction testing. The NB classifier is employed to classify data into different classes based on features. The presented technique achieves 90% and 100% accuracy on the Water Tank and Gas Pipeline datasets when NB and AE techniques are combined with feed-forward and recurrent networks

CPSs have become popular in the medical field. The combination of AI techniques and CPS in the medical field has enabled researchers to diagnose and classify common diseases. For example, Verma et al. [70] presented an IoT-cloud-fog-based CPS technique that helps identify Ulcerative Colitis (UC) disease and classify its stage. The proposed technique deploys the NB classifier on the fog layer, which acts as a classification component. The NB classifier considers various parameters, such as the health symptoms of patients and the stage of UC. Based on the mentioned parameters, the NB classifier provides information about the intensity of the disease and generates an alert if the patient is in a critical condition. The performance of the NB classifier is compared with other state-of-the-art ML techniques, i.e., kNN, SVM, and DT, for UC stage classification. The results show that the NB classifier outperforms the other state-of-the-art ML techniques by achieving 93.23% accuracy. The real-time alert generation component improves the overall efficiency of the proposed framework.

4) DECISION TREE (DT)

DT is a supervised learning algorithm for solving classification problems. The DT classifier has a tree-like structure containing internal nodes, branches, and leaf nodes. In the tree-structured classifier, the internal nodes and branches represent the features of the data set and the decision rules respectively. Furthermore, the leaf nodes show the outcome of the decision. In Industrial AIoT applications, the DT has been used to solve complex problems, i.e., edge weight prediction.

Industrial applications have also recently become part of social networks. A social network comprises various nodes, and each node could be a machine or user. In such a network, the nodes can communicate with each other. Based on the communication, the nodes can provide the ratings. The ratings are beneficial for making the directed weighted networks (DWN). Qiu et al. [71] presented the directed edge weight prediction model (DEWP) for Industrial AIoT using a DT ensemble. A directed edge is created between the nodes, and weights are assigned according to their ratings. The accuracy of the proposed model is evaluated using the Bitcoin OTC and Bitcoin Alpha datasets. The proposed model demonstrates the highest accuracy as compared to the state-of-the-art techniques.

Sharma et al. [72] presented the IDS for RPL(Routing Protocol for Low-Power and Lossy Networks)-based CPS. The proposed study presented the complete design process for IDS starting from data collection to the training of the ML models. The authors used the five popular ML algorithms, including DT to build the IDS model. The experimental results show that the DT effectively predicts the DIS and decreased rank attack (DRA) when the model is built using the embedded method. Furthermore, it is also observed that when the model is developed using correlation methods, DT also detects the Increased Version Attack.

Similarly, Safari et al. [73] have proposed the IDS for ICSs where the intrusion is detected based on the behavior of the rotating machine. Data is an essential asset in any control system. The proposed technique uses the behavior of the rotating machine and ML techniques to detect data tampering and manipulation. The behavioral model created in the proposed research produced the prediction data that is compared with the online data to detect any abnormality when the rotating machine is operating. The experimental results demonstrate that the DT-based behavioral model achieves 99.9% accuracy. Similarly, Aldehari et al. [74] compared the various ML techniques for network security because that has been useful for identifying network attacks. Experiments were conducted on three networking datasets to find the best ML technique. The results demonstrate that the DT is the most efficient and fastest traditional ML algorithm for identifying network attacks, i.e., SYN flood attacks, smurfing, and DDoS attacks.

In the medical field, IoT has enabled smart health services. Using various heterogeneous IoT and wearable devices in the medical field, we can easily monitor the patient's health and prevent critical health situations. On the other hand, the advancement in the smart healthcare system (SHS) also gives rise to many potential security concerns, i.e., the attackers can steal patient information and change the medical device's vital sign information and behavior. To prevent all types of possible cyber attacks in SHS, Newaz et al. [75] presented an ML-oriented framework for SHS named HeathGuard. The proposed framework used four different ML approaches, including DT for detecting malicious activities. The authors trained the proposed model on twelve different benign activities, including seven activities performed by normal users and five by diseased users. The evaluation results show that the DT achieves 93.1% and 90.9% accuracy in detecting benign and malicious activities, respectively.

In healthcare, fog cloud-based CPS is widely used for the early detection and identification of diseases. Mosquito-Borne Diseases (MBDs) are fatal diseases initially transmitted from the infected mosquitoes to humans and then propagate from the infected person to the other persons resulting in several fatalities [76]. Sood and Mahajan [77] presented the fog-cloud-based system that enables early detection and monitoring of MBDs to control their spread. The proposed system used the DT classifier to classify the infection category for each user intelligently. In case of any abnormality and critical situation, the fog layer generates the notifications and sends them to the user's device. A simulation experiment evaluates the accuracy and efficiency of the proposed technique. The experiment results demonstrate that the DT achieves 94% classification accuracy in classifying the MBDs.

The fourth industrial revolution (Industry 4.0) has significantly increased the productivity of the manufacturing industry. The advent of technology has led to the replacement of traditional machine maintenance by PdM in industries.

ML techniques can help to prevent any type of damage and loss of productivity. In Industrial AIoT, condition monitoring systems (CMSs) are typically used to perform PdM. Cakir et al. [78] created a CMS for PdM based on ML techniques. The proposed CMS collects real-time machine data, monitors it, and notifies the administrator if a threshold value is reached. The proposed technique used different popular ML methods for data analysis, including SVM, linear discrimination analysis (LDA), RF, DT, and kNN. It is observed that the DT is the most suitable choice for determining the threshold values of machines. Furthermore, the DT also gives fast classification results.

Data is a key enabler in the PdM of machines in Industry 4.0. ML enabled PdM helps in achieving the maximum productivity, efficiency, and equipment life. In any PdM decision, it is imperative to assess context-based information, data quality, failure history, and maintenance costs. In this regard, Arena et al. [79] presented a DT-based decision support system for PdM. The PdM policy defines many different maintenance scenarios. The DT allows us to implement the various scenarios defined in the PdM policy. The DT model uses the classification and regression trees algorithm. Modern Industry 4.0 standards help companies achieve customer satisfaction and business value. However, small companies are not able to adopt digital transformation so quickly and optimize their processes because they have a limited amount of financial resources. To overcome this problem, Kiangala and Wang [80] presented an adaptive framework using XGBoost and RF for the small manufacturing industries. Customers can search for a desired product and then see the desired customized products.

The combination of edge/fog computing and ML promotes edge intelligence. There are numerous potential scenarios of edge intelligence in the future for Industrial AIoT [27], [81], [82], [83]. In contrast, edge-based systems that process industrial data locally are more likely to be attacked. Therefore, it is critical to maintain the security of an edge-based system. However, ML-based IDSs provide sufficient security to the edge nodes. Cui et al. [84] presented an ML-driven IDS for edge computing. The proposed study discussed an optimization scheme for intrusion detection using gradient based DT (GDBT) to improve intrusion detection efficiency in edge computing systems. The experimental results show that the optimized GDBT technique increases intrusion detection efficiency. Similarly, fog computing facilitates efficient and intelligent data processing in the industrial environment. The existing frameworks and approaches are not intelligent and capable enough to process the real-time large scale data in the CPS manufacturing industry. Wu et al. [85] presented a wireless sensor network (WSN), cloud computing, fog computing and ML based framework to process the real-time industrial data for process monitoring and fault detection. The method discussed above used the RF algorithm for process monitoring and fault detection.

5) RANDOM FOREST (RF)

RF is a well-known supervised ML algorithm used to solve classification and regression problems [86]. The RF classifier is a collection of individual decision trees that work as an ensemble. An ensemble is a collection of various models to make a better prediction. In an ensemble, the prediction is not obtained through an individual model. The RF classifier can handle both categorical and continuous variables in regression and classification problems. The RF classifier has been used for solving complex classification and regression problems. In Industrial AIoT, RF is widely used to detect faults.

IoT is a key component of Industry 4.0, which advances and modernizes the CPS. However, using IoT in CPS and Industry 4.0 creates a potential cybersecurity threat and leads to cyberattacks. As a result, classic ML techniques are mainly applied in Industrial AIoT to minimize cyberattack threats and provide a fully integrated solution. RF has been proven to be an efficient technique for solving fault diagnosis-related problems and has been widely employed in the Industrial AIoT environment to construct reliable and secure security models. Tran et al. [87] presented a secure model for online monitoring of the state of the induction motor. The proposed model monitors the performance of the motor through communication channels and uses a low-cost and effort-oriented internet connection for connecting to multiple networks. The motor data is processed and visualized through the CONTACT element platform using ML techniques and depicted on the dashboard. When a cyber-attack signal is identified, it is marked as fake data and detected on the dashboard. The proposed IoT model and architecture have been validated by applying different ML techniques. Experimental results indicate that the proposed IoT model and architecture with RF algorithm achieves 99.03% accuracy for detecting motor faults. However, the proposed model is only useful for determining the induction motor status and is not applicable to other types of machines.

The constant development of novel types of malware has posed a significant threat to the Industrial AIoT [88]. Therefore, with the emergence of new types of malware, it has become difficult to identify them through malicious traffic. Traditional malware detection systems fail to detect malware in encrypted and imbalanced traffic streams. To predict the new type of malware in Industrial AIoT, Niu et al. [88] presented an improved adaptive RF (IARF) based approach that detects the different families of malware by analyzing imbalanced, drift and encrypted traffic. Adaptive RF is a modified version of the original RF algorithm that adapts to multiple types of data streams with the same hyperparameters. The experiments are conducted to evaluate the effectiveness of the proposed IARF approach using the mixed dataset. The dataset is formed by mixing the data from various sources, i.e., malware-traffic-analysis.net, CTU-13, Lastline Inc, and Malware Capture Facility Project (MCFP) datasets [89]. The proposed IARF approach is compared with the state-of-the-art techniques. The experimental results

show that the IARF approach outperforms the state-of-the-art techniques by achieving the 99.66% F1-score for classifying malware families.

Safety, efficiency, and increased production have been the need of the hour in the Industrial AIoT. The advancement of technology in the industrial environment has made it possible to monitor the health status of equipment and predict faults before critical situations occur. Cloud computing has been used in Industrial AIoT for a long time to process industrial data. Intelligent models are used to predict faults, and they are deployed in the cloud and trained with the vast amount of data. Later on, the machines transmit the real data to the cloud for processing. However, due to the increase in the number of sensors and machines in the modern industry, cloud computing faces latency and network usage problems. Therefore, the cloud is now being replaced with edge computing in Industrial AIoT for efficient and fast processing of data. Liu et al. [90] used RF-based fault detection to predict faults in an edge-based industrial environment. In the proposed study, the two algorithms are described for finding each fault type and the Minimal Edge-PLC sets for all fault types. Simulations are conducted on real-time data to evaluate the effectiveness and efficiency of the proposed model. The RF-based model is compared with the LR model and the results demonstrate that the RF-based model achieves more accuracy.

IoT is applied to a variety of utility-oriented smart solutions, including smart meters. As smart cities become more integrated, electric smart meters (ESM) are being deployed at a rapid pace. In the ESM, faults are diagnosed after the customer places a service or complaint call. Otherwise, finding a defect in the system is very difficult. Therefore, an effective ESM management system is needed. Accordingly, Siryani et al. [91] proposed an ML-based decision support system to improve the various IoT oriented ESM functions. The authors compared the proposed model with different ML techniques including DT, RF, and NB. Experimental results show that the RF achieves 96.69% accuracy for analyzing ESM data.

For decades, AI has been used in many fields. However, AI (specifically ML) has recently emerged in cybersecurity and CPS. Perrono et al. [92] applied ML to identify the critical threats in the CPS. The proposed work compares the five popular state-of-the-art ML algorithms for threat recognition including the DT. The performance of the algorithms is compared using two real-time datasets, i.e., the anomalies act dataset [93] and the water distribution dataset [94]. Experimental results show that DT outperforms state-of-the-art algorithms, such as kNN, SVM, NB, and DT. Similarly, with the rise of AI in the medical industry, ML techniques are increasingly being employed to identify and diagnose diseases. For instance, Singh et al. [95] presented an AI-based approach for the identification and classification of the Chikungunya disease using RF. In the proposed technique, the RF is used to build a model to classify Chikungunya based on severity.

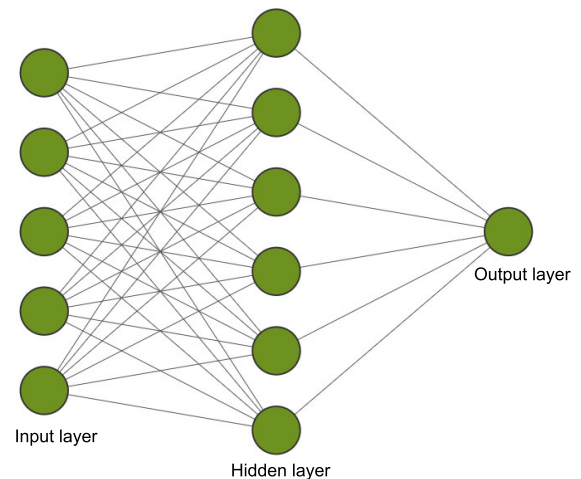


FIGURE 4. Generic layer-wise architecture of artificial neural network (ANN).

Quality control plays a vital role in the expansion and revenue of any potential business. Therefore, predicting and ensuring the quality of the products in smart factories is an imperative task. The Industrial AIoT has dramatically increased the production and quality of products by automating many processes. Early prediction of the quality of manufactured or assembled products helps in improving the industrial processes to make them better. In this regard, Feng et al. [96] presented an intelligent framework for predicting the quality of the assembled products in the Industrial AIoT environment using edge computing. The proposed intelligent framework uses the RF algorithm to predict and assess the quality of the critical features on the assembly line. According to this framework, the RF algorithm utilizes the bagging strategy, which obtains the data and creates several decision trees. The prediction is based on the results and voting of various decision trees. Likewise, Liu et al. [97] presented a privacy protection resource trading methodology for the cloud and edge PLC-enabled IIoT environment. The proposed methodology uses RF to determine the anomalous participants.

6) ARTIFICIAL NEURAL NETWORK (ANN)

ANN is one of the most widely used ML algorithms in industrial applications [30]. It is inspired by the human brain and mimics its behavior. There are three layers in an ANN model: input, hidden, and output. An input layer takes the information and passes it to the hidden layer, which performs the computation. Furthermore, the results are acquired by the output layer. Fig. 4 depicts the generic layer-wise architecture of ANN.

ANN is mainly used in Industrial AIoT applications for cyber-attack detection and prevention. DC microgrids are critical CPS that can operate independently of the primary power grid and generate their own DC power. The

infrastructure of the DC power grids makes them vulnerable to various kinds of cyberattacks, particularly false data injection attacks (FDIA). FDIA's are performed to inject false values and manipulate data to change the control of the underlying system. Habibi et al. [98] presented an ANN-based approach to detect and mitigate cyberattacks, especially FDIAs in the DC power microgrids. The effectiveness of the proposed approach is evaluated in Matlab, and the results show that the discussed approach is highly suitable for calculating the false data value and mitigating the FDIA attack.

Furthermore, the advent of 5G opens up the possibility of multiple cyberattacks, such as black/grey holes and massive DDoS attacks in the smart CPS infrastructure that can cause disruption. Krundyshev and Kalinin [99] presented an ANN-based technique for cyberattack detection in the smart CPS smart infrastructure. The proposed ANN-based approach achieves 96% accuracy for detecting all types of routing attacks in the smart CPS infrastructure.

Price-based demand response (PBDR) is being implemented in the industrial sector to balance energy demand according to price and time. As a result, it allows the industrial environment to best manage its energy consumption and prices in accordance with the needs of the market. Huang et al. [100] presented a demand response-based scheme for real-time price using ANN. The proposed ANN model consists of four layers, i.e., one input layer, two hidden layers, and an output layer. The ANN model predicts the future price of energy in the industrial environment.

Digital twin [101] is an emerging concept in Industry 4.0 that creates a virtual copy of the real system and connects the real and virtual ones together to collect, analyze, and simulate data in the virtual model. The virtual model can be used to analyze and improve the performance of the real system. It is straightforward and efficient to diagnose and monitor faults with the DT. Amini et al. [102] presented a 3-D neural network model for the wind turbine to determine the state of health of different models. The proposed system uses the ANN model with a single hidden layer to process the six sensor's data with eight channels. The authors developed a prototype for evaluating the effectiveness of the proposed system and achieved 83.33% accuracy for vibration-based condition monitoring.

IoT and CPS are key technologies involved in Industry 4.0 that guarantee benefits to the manufacturing industry. CPS is a core technology with applications in many fields, including transportation and logistics. The integration of CPS into logistics minimizes production time and operational costs. Several manufacturing companies offer logistic equipment that can be integrated with CPS technology. Nevertheless, there is no proper and effective method to evaluate the integration of such devices with CPS technologies. As a result, Matana et al. [103] presented an approach for evaluating the level of integration of logistics equipment's with the CPS technologies in Industry 4.0. The proposed method collects data from the logistics equipment and uses

the ANN to check equipment adherence with the CPS technologies. The proposed ANN-based technique showed 2% errors for assessing the adherence of logistic equipment with the CPS technologies.

Industry 4.0 integrates various technologies in the production environment to optimize production processes. These technologies include Augmented Reality (AR), sensors, actuators, and different wearable devices. All these technologies are used to optimize the processes by increasing efficiency or detecting defective processes. Identifying and timely correcting anomalies or defective processes in the industrial environment maximizes production in the industrial environment. Therefore, it is essential to identify defective processes. Accordingly, Kucukoglu et al. [104] discussed an ANN-based technique for identifying the correct and defective processes in the connector assembly tasks in an automotive-based company. The authors developed a digital glove using smart wearable technology that distinguishes between correct and defective processes. The digital glove calculates the vibration and force values on the fingers to classify the connector assembly processes. Furthermore, the proposed technique uses the two-layer feed-forward ANN model to classify the signals from the gloves into correct or defective processes. According to experimental results, the proposed ANN model can accurately predict the correctness or defect of a process 95% of the time. Experimental results indicate that the proposed ANN model is 95% accurate in predicting whether a process is correct or defective.

7) FEDERATED LEARNING (FL)

FL is an emerging branch of AI in which ML models are trained on various decentralized devices such as mobile phones, edge devices, or other smart devices [105]. FL enhances the security and privacy of data, as each device or worker participating in FL is trained on local data. In FL, the fundamental concept involves training ML models on decentralized, local data sources. These models are typically stored on individual devices like smartphones, gateway devices, or laptops. Periodically, these local nodes exchange model parameters with a central server. This iterative process facilitates the aggregation of these parameters to construct a more robust and comprehensive global ML model. The training data resides on individual nodes, typically decentralized devices, thereby ensuring data privacy. Only model parameters are transferred to a centralized cloud storage system. A global ML model is hosted on a central cloud server. Upon receiving model parameters from individual devices, this central server aggregates them by averaging to generate a unified global model. As a result, individual nodes engage in collaborative learning via a shared model. Among the advantages of FL over traditional ML models are data privacy, reduced latency, and lower power consumption. FL not only protects sensitive user information by keeping it localized but also offers the benefit of personalized ML models, thereby improving the user experience.

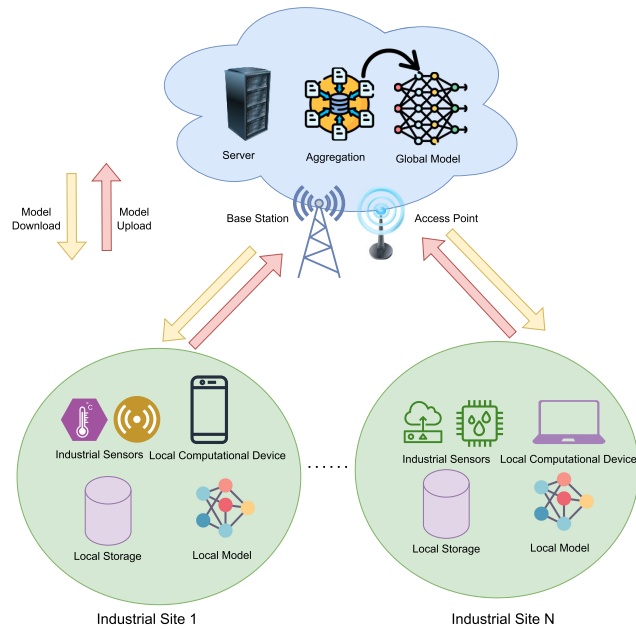


FIGURE 5. An overview of FL in Industrial AIoT enabled systems.

In an Industrial AIoT environment, FL can be integrated for various purposes, including security and data privacy, reduced network latency, scalability, and improved performance. Security and data privacy are enhanced as sensitive data remains local, reducing exposure to potential cyberattacks. FL significantly reduces network latency and communication costs, particularly when IIoT devices generate large volumes of heterogeneous data. By processing this diverse data locally and sending only summarized results to the cloud, FL streamlines data management and lowers associated overheads. Improved network performance arises from localized processing, which reduces overall network traffic. Scalability is facilitated as FL accommodates the large and diverse data volumes generated by IIoT devices, allowing central algorithms to learn from summarized insights rather than raw data. These benefits make FL an optimal choice for managing the complexities and risks associated with Industrial AIoT applications. Fig. 5 shows an overview of FL in Industrial AIoT enabled systems.

In distributed Industrial AIoT environments, the lack of efficient security mechanisms makes industrial devices susceptible to various types of attacks. These vulnerabilities can expose sensitive data to malicious users, posing significant threats to industrial applications. Given the urgency of ensuring data reliability, security, and privacy, anomaly detection has emerged as a key solution. Several FL-based frameworks have been proposed to address this issue. Wang et al. [106] introduced Federated DRL Empowered Anomaly Detection (FLAD), which utilizes DRL and the Deep Deterministic Policy Gradient (DDPG) algorithm to detect anomalies at both local and global levels. Experimental results confirm FLAD's effectiveness in ensuring privacy, reducing latency, and increasing throughput. Sim-

ilarly, Yazdinejad et al. [107] proposed a framework for enhancing security in blockchain-based IIoT networks. The authors presented an FL-based framework called "Block Hunter," designed to automatically detect anomalies and cyber threats in blockchain-based IIoT networks, particularly in smart factories. The framework employs a cluster-based architecture and utilizes various ML models, such as the Neural Encoder–Decoder (NED) model, Isolation Forest, K-Means, etc., to identify anomalous activities while preserving data privacy. Liu et al. [108] further extended the FL-based anomaly detection approach by introducing an Attention Mechanism-based CNN-LSTM model for time-series data. In the context of ICS, Huang et al. [109] proposed FedEx, a framework that combines Variational AE (VAE), FL, Support Vector Data Description (SVDD), and Explainable AI (XAI) for efficient and interpretable anomaly detection. Additionally, Ma et al. [110] introduced FedAnomaly for cloud manufacturing, which employs Transformer models and Multilayer Perceptrons (MLPs) to detect anomalies while maintaining data privacy. Each of these frameworks aims to enhance the state of anomaly detection in various industrial settings.

As Industrial AIoT grows, FL is becoming increasingly important for ensuring data safety and privacy. Most state-of-the-art research focuses on this aspect. For instance, Bugshan et al. [111] presented an FL-based DL framework aimed at preserving privacy in IIoT. This proposed framework utilizes both edge and cloud computing to establish a service-oriented architecture for Federated DL (FDL) with differential privacy. It specifically employs residual networks to generate trustworthy, locally-trained models while maintaining data privacy. Experiments were conducted on the brain tumor (BT) dataset [112] using the popular CNN variant, ResNet50, to enhance BT detection accuracy. Makkar et al. [113] proposed an FL-based approach named "SecureIIoT" for maintaining the privacy of industrial data by detecting malicious industrial devices. The authors employed Paillier encryption and conducted tests using the XIIoTID dataset. This proposed framework shows promising results in terms of accuracy, detection rate, and efficiency. Similarly, Wei et al. [114] proposed a novel chameleon hash scheme with a changeable trapdoor (CHCT) to secure FL in IIoT settings. The scheme allows for the secure and efficient storage and management of data and addresses key exposure and trapdoor abuse issues. The experimental evaluations demonstrate the practical utility of CHCT in terms of accuracy and efficiency.

The industrial AIoT environment is resource-constrained; therefore, selecting appropriate communication and computing resources is crucial for effectively training the FL model and performing tasks. To tackle this issue, the authors in [115] proposed a scheme that optimizes device selection and bandwidth allocation. Their aim is to minimize time-averaged costs while adhering to long-term energy budgets and delay constraints. They employed Lyapunov optimization theory to solve this long-term optimization problem and

TABLE 3. Summary of ML techniques for industrial AIoT.

Technique	Description	Main Contribution in Industrial AIoT	Limitations
K-Nearest Neighbor (K-NN)	Suitable for both regression and classification. Assumes data similarity for categorization.	- Defect identification [41] - Anomaly detection [45]	Inefficient with large datasets due to high computational cost.
Support Vector Machine (SVM)	Classical ML algorithm for classification and regression. Uses hyperplanes for classification.	- Agriculture data analysis [51] - Fault diagnosis [52] - Security in Industrial AIoT [54] - Anomaly detection [55] - Human activity identification [56]	- Not suitable for large datasets - Performance degrades with noisy data.
Naive Bayes (NB)	Probabilistic ML for classification. Assumes feature independence.	- Intelligent fault detection [66] - Intrusion detection [67] - Disease classification in healthcare [70]	Assumes feature independence, limiting its ability to capture feature relationships.
Decision Tree (DT)	Supervised learning for classification and regression. Good for complex problems.	- Complex problem solving [71] - Intrusion detection [73] - Disease classification [77] - Predictive maintenance (PdM) [79]	- Introduces increased model complexity - Longer model training time - Sensitive to small dataset changes, leading to variations in the decision tree structure.
Random Forest (RF)	Ensemble ML for classification and regression. Combines decision trees.	- Fault diagnosis [87] - Malware detection [88] - Industrial AIoT support [91] - Disease classification [95] - Quality control [96]	Complexity and slowness increase with a higher number of trees.
Artificial Neural Network (ANN)	Inspired by the human brain. Consists of input, hidden, and output layers.	- Cyber-attack detection [99] - Demand response strategy [100] - Defect identification [104]	- Requires a large amount of data for training - High computational resources
Federated Learning (FL)	Decentralized ML for privacy and security.	- Anomaly detection [106] - Privacy and security [107] - Resource optimization [115]	- Challenging to train different models on diverse devices - Varying computational capabilities across devices can slow down the learning process - High bandwidth requirements for model updates between edge devices and central server

designed an iterative algorithm for round-wise decision-making. Extensive experiments validate that their scheme outperforms other methods in cost-efficiency while ensuring FL training performance. In industrial settings where devices are densely deployed, an increase in the number of clients directly leads to elevated training latency. To mitigate this, resource allocation becomes essential for minimizing training time and ensuring timely computation. In this context, Gao et al. [116] proposed a resource allocation scheme called RaFed, designed to minimize FL training latency. RaFed takes into account both computation and communication latencies, formulating the issue as an NP-hard optimization problem. A heuristic algorithm is proposed to iteratively improve training latency, and experimental results indicate that RaFed reduces FL latency by 29.9% compared to state-of-the-art methods.

While we have aimed to mention some of the most prominent studies focusing on the important aspects of FL in Industrial AIoT, it is important to note that this is a rapidly evolving field. Therefore, we have selected some studies to highlight the importance of FL in the industrial AIoT environment.

B. DEEP LEARNING

In this section, we present various state-of-the-art DL techniques used in Industrial AIoT.

1) AUTOENCODER (AE)

An AE is an unsupervised neural network that consists of two parts: an encoder and a decoder. A hidden layer called the code connects the encoder and decoder. The encoder takes input, compresses it, and generates the code. Furthermore, the decoder gets the code and reproduces the input. Since AE is an unsupervised learning method, it does not require explicit label data for training. The encoder and decoder could be feedforward neural networks, consisting of convolutional and pooling layers, or they could be fully connected [24].

With the expansion of the manufacturing industry, the deployment of mobile devices (MD) is also increasing in Industrial AIoT. The MDs continuously produce and transmit data, but the data can contain noise due to any disturbance or hindrance. Thus, data transmission should be guaranteed from the relay nodes to the destination in this situation. Rui et al. [117] presented a link prediction strategy that used the stacked denoising AE (SDAE). The proposed technique maintains historical link information and predicts MD links. The SDAE is a collection of denoising AEs primarily used to remove noise or denoise corrupt information [118]. The proposed link prediction technique is compared with state-of-the-art methods, i.e., kNN, SVM, and Katz. The experimental results show that the SDAE-based technique surpasses existing techniques in terms of accuracy, precision, and recall.

Security has always been a concern in Industrial AIoT. As a result of the new threats, security has become a potential

concern in Industrial AIoT. Researchers have applied ML and DL techniques to solve security threats in Industrial AIoT. AE plays an important role in minimizing security threats. Aouedi et al. [119] proposed a FL-based approach for attack detection using unlabeled data to train the AE on each device. The AE learns the low-dimensional features in the training phase. Additionally, Khan et al. [120] proposed an AE-based framework for threat detection.

DL-based models have been found efficient for anomaly detection. They have shown satisfactory performance in rule training, adaptability, and anomaly detection accuracy. Different AE variants, such as variational AE (VAE) and conditional AE (CVAE), are useful for building DL-based IDSs. Liu et al. [121] presented an IDS for Industrial AIoT that works on the data level. Three different data-related techniques are presented to solve the data imbalance and augmentation problems. In data augmentation, new data points are generated from existing data points, increasing the amount of data. As part of the proposed IDS, the data augmentation scheme is based on VAE, while CVAE is used for data balancing. Furthermore, in the third step, data balancing is also based on under-sampling. All three steps are combined in a DL-based IDS. Moreover, Sun et al. [122] presented an AE-based intelligent framework for collecting data from heterogeneous devices and intelligently joining it for various purposes.

2) CONVOLUTIONAL NEURAL NETWORK (CNN)

Among the plethora of CV techniques, CNN is one of the finest and most efficient techniques to solve complex problems [123], [124], [125], [126], [127]. Furthermore, CNN has shown great performance in natural language processing (NLP), recommendation, and surveillance systems [128], [129], [130]. A convolutional network typically consists of convolutional, pooling, and fully connected layers [131]. Fig. 6 shows the generic architecture of a CNN.

CNN has aided Industrial AIoT in automating industrial operations and anomaly detection. The CNN model requires high computational resources for fast and efficient performance. Deploying CNN on resource-constrained devices can create a performance hindrance. However, in the industrial sector, conventional cloud-based applications suffer from potential challenges, i.e., latency and bandwidth usage. Therefore, using edge computing and migration of CNN to edge nodes could be helpful to increase efficiency and performance in Industrial AIoT [132]. Yang et al. [133] presented the transfer learning-based security-oriented edge CNN framework for a 5G industrial edge network. The CNN model was deployed and trained on the edge server. The proposed CNN framework reduced latency and energy consumption in 5G industrial networks. The experiments were conducted using the ImageNet dataset, showing that the CNN-based model achieves 85% accuracy. Similarly, Wang et al. [134] proposed a cloud-edge-oriented collaborative framework based on CNN. The proposed CNN-based

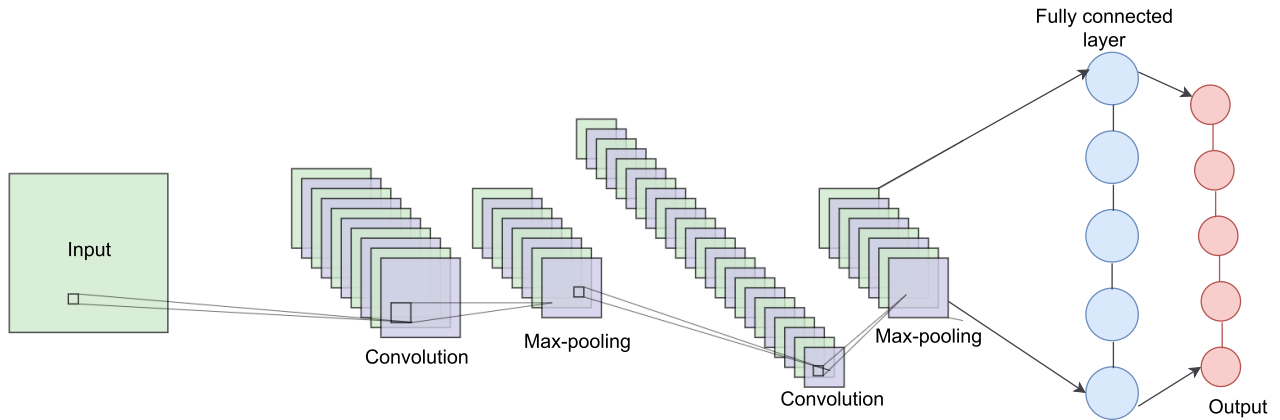


FIGURE 6. Architecture of convolutional neural network (CNN).

framework ensures privacy preservation in a Cloud-Edge collaborative environment.

In Industrial AIoT, communication between smart devices is essential for sharing data. However, inter-device communication compels many potential threats and cyber-attacks. These types of attacks can lead to the worst safety-related situations. Nedeljkovic et al. [135] discussed a CNN-based model to detect cyber attacks on communication links among various devices.

Efficient resource allocation and data security are two key concerns in Industrial AIoT. Due to the increased number of nodes in the industrial environment, interference can potentially cause network resource loss and give rise to security problems. To address these issues, Goswami et al. [136] presented a CNN-driven methodology for efficient resource allocation and enhanced security. The CNN-based model extracts feature values and calculates the signal-to-inference ratio (SIR) for resource allocation. Simulated and experimental results were used to evaluate the effectiveness of the proposed model. The experimental results demonstrate that the CNN-based methodology outperforms state-of-the-art techniques. Furthermore, Ding et al. [137] introduced an adversarial defense mechanism and utilized CNN for classification to assign images to specific classes.

Recommender systems [138] are great tools for personalized recommendations. The use of a CNN-based recommendation system is increasing day by day and has also been found beneficial in the robust Industrial AIoT environment. For instance, the recommender system can mine data, analyze it, and provide valuable recommendations to end-users. However, recommender systems often face the sparsity problem, which makes it difficult to find a large amount of similar data. Therefore, they can be vulnerable to malicious recommendation attacks. To address this issue, Li et al. [139] presented a CNN-based model that employs a personalized attentional approach to mitigate the sparsity problem and enhance recommender system performance. The proposed model uses both word-level and review-level

attention models with CNN, giving high attention to critical words and informative reviews.

Intelligence is an integral part of Industrial AIoT for performing maintenance and making operations optimal and autonomous. AI methods, particularly DL, have been useful for fault diagnosis in industrial equipment. PdM and early diagnosis can save many resources in the industrial environment. Intelligent multi-sensing technology and CNN in an industrial environment allow us to gather information from various sources and process it efficiently. Accordingly, Wang et al. [140] presented a technique to perform the fault classification based on vibrational signal propagation using CNN. Firstly, the proposed method takes the vibrational signals, applies time series, and converts the raw signal into an image because it is difficult to process the raw signal. Secondly, the CNN model is trained to classify the signal-based images. The described technique is evaluated on three different datasets and compared with the state-of-the-art techniques. The experimental results show that the proposed method achieves 99.92%, 99.68%, and 99.25% on three different datasets, which is higher than the state-of-the-art DL approaches. The results indicate that CNN is one of the most efficient DL approaches for image-based classification. Similarly, Pham et al. [141] used CNN to perform the image-based classification to classify the tissue state, i.e., normal and scar tissue. In addition, CNN is widely used in agriculture to detect diseases. Udalapally et al. [142] also used the CNN model to analyze the crop images and classify them into healthy and diseased plants. The proposed CNN-based framework achieves 99.24% accuracy for plant disease prediction.

Besides image-based classification and prediction, CNN has also proved to be a very effective tool for video surveillance systems. Video surveillance systems have become very popular in cyber-physical-based infrastructure. They have been widely used for security reasons, including academic institutions and university campuses. Many video surveillance systems have been proposed and developed for

numerous domains, and the availability of activity datasets is ensured. However, a specific video surveillance and activity recognition system for academic activity recognition has not been proposed. Therefore, Wasim et al. [143] proposed a CNN-based academic activity recognition model for a cyber-physical-oriented smart campus. The authors also created a novel campus activities dataset, and the CNN model is trained on that dataset. The proposed CNN model consists of four convolutional and max-pooling layers, two fully connected layers, and one classification layer. Experiments were performed to evaluate the proposed model's performance regarding accuracy, memory consumption, and execution time. The results demonstrated that the proposed CNN-based model for academic activity recognition achieves 98% accuracy.

Tang et al. [144] presented a CNN model to recognize human activities. Furthermore, drowsiness can be a major factor in tragic road accidents. Therefore, an intelligent system for drowsiness detection has the potential to minimize road accidents and determine the driver's vigilance status. Consequently, Tamani et al. [145] discussed a CNN model that determines the vigilance status of the driver based on real-time facial expressions. The authors evaluated the proposed system using the University of Texas at Arlington-Real-Life Drowsiness Dataset (UTA-RLDD) [146]. The experimental results showed that the proposed system achieved 96.8% accuracy when processing 8.4 frames per second. The study discussed in [147] recognized hand gestures, and CNN was used to classify the hand gestures. Liu et al. [148] presented an approach to recognize the manufacturing process and used the CNN model to retrieve low-dimensional features, which were then fine-tuned to identify specific tasks in a process.

In an Industry 4.0 environment, humans, robots, and various mechanical machines perform a range of tasks. Therefore, it is essential to detect humans in indoor environments and distinguish them from machines to avoid accidents and collisions. Khan et al. [149] presented an approach for detecting and localizing humans in indoor Industry 4.0 environments. The proposed approach used CNN to extract patterns from real-time data and classify objects as either machines or humans. Experiments were conducted in two indoor environments to assess the performance of the proposed system.

Coronavirus disease (Covid-19) hit the world in 2019 and became a pandemic. It continues to have devastating social and economic effects all around the globe [150]. Industry 4.0, a promising domain, has the potential to fulfill society's manufacturing and economic needs during the pandemic crisis. The prominent components of Industry 4.0 are IoT, AI (ML&DL), and digital technologies. For example, X-ray images and computed tomography (CT) are used to detect Covid-19 in a person. Therefore, Industry 4.0 is beneficial for controlling the Covid-19 outbreak. Ahmad et al. [151] presented an Industry 4.0 oriented framework to control the spread of Covid-19 using CNN. The proposed CNN-based

model extracted the X-Ray features and classified the patient into infected or disinfected categories. The proposed model was compared with futuristic models, i.e., VGG16, AlexNet, and InceptionV3. The experimental results demonstrated that the proposed CNN model outperformed the state-of-the-art models and achieved higher accuracy.

3) RECURRENT NEURAL NETWORK (RNN)

RNN is a type of ANN that is best suited for processing time-series or sequential data. The inputs are in the form of a sequence of words, events, or data. The computations in RNN are related and dependent on each other. The computation performed at a specific timestamp is based on the computation of the previous timestamp. For instance, in a sentence, the prediction of the next word depends on the previous word. Typically, an RNN consists of an input state, a recurrent state responsible for performing all computations, and an output state that delivers the result or output [152], [153], [154].

RNN has been a part of Industrial AIoT for various tasks, including modeling, PdM, identifying malicious patterns, and fault diagnosis [155], [156], [157]. For instance, Kumar et al. [158] used RNN in IIoT to identify malicious patterns. Woźniak et al. [159] presented an RNN-based model for malware threat detection in CPS. The model performs network traffic analysis. The network traffic information is captured at the firewall level and passed to the RNN model as input. The proposed RNN model assesses the information for malware threat detection and sends the results to the firewall to take any necessary security-related actions. The experimental results show that the proposed model achieves 99.9957% accuracy for malware threat detection. Moreover, Habibi et al. [160] used RNN to detect FDIA's in DC microgrids.

Sensors are widely used in Industrial AIoT applications to determine the asset's current state and detect any changes. Similarly, sensors acquire the vibration pattern from the asset to diagnose the state and predict the asset's future failure. Kizito et al. [161] presented a Long Short-Term Memory (LSTM) based RNN model for finding defects and determining the remaining useful life of industrial components or assets. The proposed model is trained and evaluated on the motor degradation dataset. In the case of failure prediction, the experimental results show that the proposed model is efficient in predicting failures and can predict a failure 24 hours in advance with 93% accuracy. Imamura et al. [162] used the RNN-based model to diagnose and predict rotating machine faults.

Various industrial equipment (sensors, machines, and robots) generate large amounts of data that can be used for anomaly detection, PdM, and process automation. Particularly in the manufacturing system, data comes in the form of sequences and is time-series data. Wang et al. [163] presented an RNN-based model to detect anomalies in time-series data. The authors trained the proposed model on data

obtained from system operations. The proposed model takes time-series data as input and reconstructs the input data in a time-series manner. The input and output are compared to detect anomalies. Additionally, Aledhari et al. [74] compared various classical ML and Neural Network algorithms for network security applications on different network-based datasets named KDDCup 99, NSL-KDD, and ADFA IDS 2017. The experimental results verified that RNN showed the highest performance on all three datasets among the neural network models.

4) RESTRICTED BOLTZMANN MACHINE (RBM)

RBM is a type of ANN used for various purposes, including dimensionality reduction, classification, regression, feature learning, etc. [164]. The RBM architecture consists of an input or visible layer and a hidden layer. The input layer receives the input, and the hidden layer processes it to extract features [165], [166]. The input and hidden layers are connected to each other. Furthermore, RBM is categorized into Deep Belief Network (DBN) and Deep Boltzmann Machine (DBM) [35].

RBM is useful in Industrial AIoT to enhance efficiency, reliability, and safety. For instance, Singh et al. [167] presented a blockchain-based security-oriented mechanism for CPS to transmit information to drones. The authors utilized the DBM technique for appropriate node selection.

In the healthcare industry, IoMT ensures the availability of various online healthcare services and patient monitoring. Healthcare applications are highly dynamic, and mobile devices are used to access data. However, the patient's private health data must be highly secured. Therefore, data privacy and execution cost are two major challenges in healthcare-oriented Industrial AIoT applications. Hence, Lakhan et al. [168] presented a security mechanism for healthcare applications using RBM. The authors compared the proposed technique with state-of-the-art mechanisms, and experimental results show that the proposed RBM-based technique outperformed existing schemes.

5) GENERATIVE ADVERSARIAL NETWORK (GAN)

GAN is the popular DL-oriented generative technique used for generating additional/synthesized data. A GAN model consists of two parts, i.e., i) a generator and ii) a discriminator [169], [170]. Both of the components in the GAN model have contradictory objectives. On the one hand, the generator component analyzes and learns from the real data to generate the synthesized data. On the other hand, the discriminator distinguishes between real and synthesized data. The discriminator sends feedback to the generator and updates the model based on this feedback. The primary purpose of this model is to generate identical and indistinguishable samples from the real data. If the discriminator successfully distinguishes between real and synthesized data, then the generator model is penalized. Besides, the discriminator is penalized if it fails to distinguish

between the real and the synthesized data [171], [172]. GAN is a useful technique to generate synthesized data and train the model to optimize the system's accuracy and efficiency in case of a lack of training data.

Intelligence has become indispensable for industrial automation applications and increased productivity. A significant amount of data is required to train the model to produce an accurate prediction. Nevertheless, one critical issue for Industrial AIoT systems is the lack of sufficient data. Despite the fact that sensors, actuators, machines, and robots generate a significant amount of data, the availability of sufficient data is a potential challenge in new Industrial AIoT systems. Therefore, it is necessary to ensure sufficient data to train the model in Industrial AIoT. To solve this issue, Qian et al. [173] presented a GAN-based framework to generate the new data from the real data for the model training. The generation of new data gives rise to potential security threats. Therefore, the authors also discussed the security risks that can arise from synthesized data and presented an approach to alleviate these types of risks. The proposed GAN model uses CNN to generate the synthesized data. The model is tested on different scenarios using industrial data to prove its effectiveness. Experimental results show that the lack of sufficient data greatly affects the training and accuracy of a model.

Furthermore, the deep CNN-based GAN (DCGAN) model generates more accurate synthesized data. Wang et al. [174] presented a DRL-based model for securing software-defined industrial applications. In software-defined industrial applications, some or all of the industrial processes can be monitored and managed through software. Since it is difficult to find the real-time dataset for Industrial AIoT, the authors used the GAN model to generate the industrial traffic data based on real data.

The limited availability of malware data in Industrial AIoT makes it challenging to train the model for efficient malware detection. Moti et al. [175] presented a framework titled MalGAN to generate the samples and detect the malware efficiently. In the proposed framework, the CNN and GAN models are integrated to detect the high-level features and generate the samples. The proposed framework is evaluated using Windows datasets. Experimental results show that the proposed framework outperforms the futuristic approaches in terms of accuracy for malware detection.

GAN is a suitable technique for solving the distribution problem for the imbalanced datasets in Industrial AIoT. The uneven distribution of the data can lead to invalid model training and false predictions. Zhou et al. [176] presented a GAN-oriented framework to solve the imbalanced data problem for Industrial AIoT. The proposed GAN-based technique augments data, and the generator component connects to a feature classifier. The feature classifier works in collaboration with the generator during the training process. It is based on the shared set of featured weights that help correct the distribution bias problem during the data augmentation process. The random label sequences of the minority classes are assigned to the generator that generates the data based

on these random samples. Consequently, the data of the minority classes can be enhanced efficiently by using the proposed framework. The proposed framework was evaluated on two real-world imbalanced datasets, i.e., NSL-KDD and UNSW-NB15. The results prove that the proposed GAN model outperforms the state-of-the-art baseline methods and GAN-based models for solving the imbalanced data and generating suitable high-quality data in the minority classes. Farajzadeh-Zanjani et al. [177] also used the GAN model to solve the class imbalance problem for faults and cyber-attacks classification in the CPSs. Taheri et al. [178] presented a router attack detection mechanism in IIoT and used GAN-classifier (GANC) to detect the attacks. The proposed GAN-C model distinguishes real data from synthesized data, identifies attacks, and classifies them. The imbalanced data of faults can sometimes interfere with the calculation of the remaining useful life (RUL). The RUL parameters are useful in PdM to predict machine faults before time. Behara and Misra [179] proposed the conditional GAN (CGAN) approach to solve the imbalanced data issue for PdM. The proposed CGAN approach learns the existing underlying data and generates additional artificial data to solve the data imbalance problem.

Integrating various cutting-edge technologies, especially cloud & fog computing, CPS, IoT, and big data enables practitioners and researchers to utilize the Industry 4.0 capabilities in the manufacturing industry. Usually, the data generated from the heterogeneous sensors in Industrial AIoT is used for fault prediction and anomaly detection. However, the missing sensor data causes reliability problems in fault prediction and classification. Dzaferagic et al. [180] studied the impact of the missing sensor's data on the monitoring system for fault detection and classification. Furthermore, the authors presented a GAN-based framework that finds the relationship between the data and generates similar data to replace the missing sensor's data. The performance of the proposed GAN model is optimized based on fault detection and classification. The model is trained and tested on the Tennessee Eastman chemical dataset [181], and the results indicate that synthesized data from the GAN model diminish the effect of missing data and improve the overall performance of the monitoring system for fault identification and classification. Similarly, Kong et al. [182] integrated the GAN with bi-directional LSTM and the attention mechanism for anomaly detection in the industrial environment.

Fog computing is a novel distributed computing paradigm that provides the cloud-like service at the edge of the network [183]. Fog nodes are geographically distributed. It has proven efficient for processing massive amounts of data with low latency and network bandwidth usage compared to cloud computing [184]. Furthermore, fog computing has received attention from researchers and has been widely used in various domains including healthcare, transportation, and Industry 4.0 [1], [15], [185]. However, fog computing could be prone to various cyber-attacks in the CPS. The ML models are proposed and trained to prevent cyber attacks in the

fog computing environment. However, with the advent of novel cyber-attacks and the absence of labeled data, it is quite challenging to identify such types of attacks. GAN is a viable unsupervised technique in such scenarios for detecting cyber attacks. However, in CPS, it is essential to detect the cyber attack with low latency before it damages the system. De Araujo-Filho et al. [186] proposed a fog-based IDS for CPS that used GAN. The proposed model ensures low latency requirements and computes the reconstruction loss. The proposed model is compared with the state-of-the-art techniques, and experiments are conducted to evaluate the detection latency & performance. The results show that the proposed model outperforms the state-of-the-art techniques in terms of low latency and performance.

FL can be used with GAN for anomaly detection in Industrial AIoT. The central server selects the suitable edge devices or servers with enough local data where we can train our model or algorithm [187]. Afterward, the edge devices or servers send the parameters and biases of the locally trained model to the central server. The model is averaged according to updates to improve the shared model. Basset et al. [188] presented an FL-based semi-supervised framework for anomaly detection in fog-assisted industrial smart grid (ISG) using GAN. The GAN model is used to generate high-quality data on minority classes. The experimental results show that the proposed framework achieves 97.28% and 96.36% accuracy for binary and multi-class classification, respectively. Taheri et al. [178] introduced an FL-based approach to detect Android malware in the industrial environment. The proposed approach is composed of a client or participant side and a server side. The GAN model generates adversarial attack data and implants it into industrial data. The Android malware is detected on the server side, and the vulnerable data sample is deleted.

6) DEEP REINFORCEMENT LEARNING (DRL)

DRL combines ANN and reinforcement learning to solve complex real-world problems. It enables software agents to learn, achieve specific goals, and solve particular problems. The agents interact with the environment, and learning improves over time [189]. It is not always possible for agents to have complete prior knowledge of the environment. If any information is missing, the agents can continually interact with the environment and collect the required information. DRL has been found quite helpful in solving complex real-time Industrial AIoT problems in transportation, healthcare, agriculture, and networking applications [190], [191], [192], [193], [194].

Mobile edge computing (MEC) is commonly used in Industrial AIoT for task offloading and processing. High-performance MEC devices can be deployed in the industry in a distributed manner to bring computational resources closer to user devices. Latency-sensitive tasks require high computational resources. Therefore, these types of tasks are partially offloaded to the MEC server, enabling tasks to be processed locally and on MEC servers simultaneously.

Deng et al. [195] presented a DRL-based intelligent scheme for latency-sensitive partial task offloading using MEC. The proposed DRL-based framework comprised Q-learning and the DDPG algorithms. The Q-learning algorithm makes offloading decisions, and DDPG handles continuous offloading decisions. The proposed Q-learning and DDPG schemes are compared with local computing, full offloading, and deep Q network. Experimental verification proves that the proposed Q-learning and DDPG schemes minimize computational delay by 23% and 30%, respectively. Moreover, Wei et al. [196] presented the locally privacy-oriented DRL technique for task offloading in the edge cloud environment. The proposed approach generated differential noise to guarantee privacy requirements.

MEC servers have fewer computational resources compared to cloud servers. Therefore, when offloading computationally intensive tasks to MEC servers, it is crucial to efficiently manage computational resources to avoid performance bottlenecks. Chen et al. [197] presented a DRL-based dynamic and efficient resource management scheme for MEC servers in Industrial AIoT. The proposed scheme used a DRL-based dynamic resource management algorithm (DDRL) to efficiently manage computing resources and transmission power. Ren et al. [198] presented a DRL-based scheme for high computation-demanding task offloading in a fog-based Industrial AIoT environment. The proposed study used the multi-agent DRL-based approach to identify and select the appropriate fog node or access point to offload the task and minimize energy consumption.

DRL is an excellent tool for performing PdM-related tasks. Ong et al. [199] presented the PdM framework for manufacturing that used DRL to predict machine health and allocate workforce resources accordingly. Almost every industrial machine is equipped with sensors that transmit the machine's data to the edge server through a gateway. This machine data is analyzed to predict the machine's health state. In the proposed study, a DRL scheme was used for scheduling maintenance-related tasks and managing resources accordingly. Furthermore, when a fault is predicted, it is necessary to assign an appropriate technician to fix it effectively. In [200], the authors discussed the DRL-based PdM approach for efficient and effective resource management.

7) TRANSFORMER

In Industrial AIoT systems, the most commonly collected type of data is time-series or sequential data, owing to the frequent transmission of signals from heterogeneous industrial devices, including sensors. Transformer-based models, following an encoder-decoder architecture, have proven effective for time-series forecasting tasks [201]. These models excel at extracting correlations within time-series data and making accurate predictions. They utilize the attention mechanism, a concept that has recently gained prominence in the field of DL [202]. Inspired by human biological systems that naturally focus on distinctive elements in large datasets,

the attention mechanism has been widely adopted across various application domains, showcasing the intersection of human cognition and technological innovation as deep neural networks continue to advance. The attention mechanism allows models to allocate computational resources more effectively to critical information. The attention mechanism is defined in Eq. (1):

$$A(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \quad (1)$$

where A , Q , K , V , and d represent attention, query, key, value, and input dimensionality, respectively. In the context of Industrial AIoT applications, predictions are made using sensor values. The attention mechanism, as described in Eq. (1), plays a crucial role in highlighting significant information over time. The encoder is designed to efficiently process longer sequences, taking these input sequences to identify dependencies within the data and map them to a higher-dimensional feature space. Before the sequences enter the encoder, positional encoding is added to capture the correct order of the sequence. The decoder takes the high-dimensional feature map from the encoder and reconstructs the output sequence, utilizing the attention mechanism to focus on relevant features and enhance predictive accuracy. Transformers employ the multi-head attention mechanism, as described in [203]. Multi-head attention serves as a specialized module and an extension of the attention mechanism, designed to execute the attention process in parallel across multiple pathways or heads. This concurrent operation allows for a more comprehensive understanding of the data by focusing on different aspects simultaneously. Finally, a fully connected layer maps the output of the decoder to the prediction results. Fig. 7 illustrates the high-level architecture of Transformer, comprising both the encoder and decoder components.

In Industrial AIoT applications, Transformer is primarily used for prediction tasks, with a focus mainly on PdM and RUL, as well as classification tasks. As previously discussed, the industrial environment comprises a variety of heterogeneous devices, leading to the generation of complex traffic data. Therefore, classifying the flow of this traffic becomes essential for gaining better insights. Zhao et al. [204] proposed a framework called Flow Transformer, based on Transformer and specifically designed for traffic classification in industrial applications. This framework employs a multi-head attention mechanism to focus on relevant flows within a sequence, thereby enhancing traffic analysis. To further improve the classifier's performance, the authors utilized an RF-based feature selection method. Experimental results from three real-world traffic datasets demonstrate that Flow Transformer outperforms existing methods in the field.

In PdM applications, a key focus is to predict the RUL of a component. The RUL is an estimate of the duration for which the component is expected to operate in accordance with its design specifications before necessitating replacement [205]. Over the past few years, DL-based methods have proven

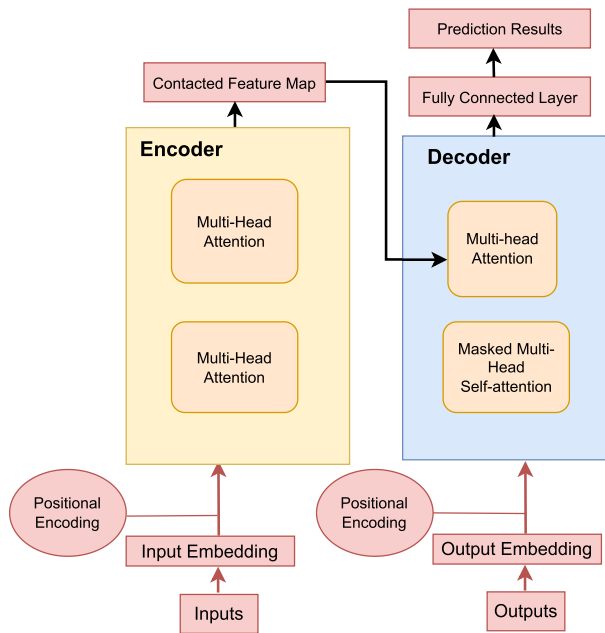


FIGURE 7. High-level architecture of transformer.

effective for determining the RUL of equipment. Specifically, Transformers equipped with the attention mechanism are capable of determining both the spatial correlation between sensor values and the degradation process, as well as the RUL values. The attention mechanism pays more attention to the sensor values that contribute more to the degradation process and helps determine the RUL. Ren et al. [206] presented a Transformer-based method for RUL prediction. The authors introduced a high-order transformer model for multiscale feature extraction, aiming to improve the accuracy and efficiency of RUL prediction. The paper further discusses tensor ring decomposition as a method to lighten the computational load of the Transformer model. Zhang et al. [207] presented a Transformer-based framework for RUL prediction, based on the attention mechanism and following the encoder-decoder architecture. The proposed framework is named Dual-Aspect Self Attention based on Transformer (DAST) and leverages two encoder layers. Both encoder layers process the sensor and time-step features in parallel. Experiments are conducted on the NASA turbofan engine Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) dataset. The experimental results demonstrate that the proposed DAST framework outperforms state-of-the-art methods in terms of better RUL prediction. Wang et al. [208] proposed a joint DL-based architecture comprised of a Transformer encoder leveraging the attention mechanism and CNN. In the proposed architecture, the encoder extracts the model's global dependencies in the input data using the attention mechanism, whereas CNN focuses on local feature extraction and dependencies. The effectiveness of the proposed architecture is evaluated using the C-MAPSS turbofan engine dataset. Similarly, Guo et al. [209] proposed a

Transformer-based prediction method with a multiscale gated CNN for rolling bearing RUL prediction. The CNN and GRU are combined for feature extraction, whereas the multi-layer Transformer model extracts the dependencies in the input data and makes the RUL prediction.

This section has outlined the prominent applications of Transformer in the field of Industrial AIoT, where they are primarily utilized for prediction-related tasks. However, they can also be employed for other tasks such as anomaly detection, process optimization, quality control, and real-time monitoring.

8) LARGE LANGUAGE MODELS (LLMs)

LLMs have received significant attention across academia and industry due to their remarkable performance across a variety of applications [213]. The LLMs exhibit versatility when it comes to tackling a wide range of problems, unlike earlier models which are designed for specific tasks. The effectiveness of LLMs extends from general natural language tasks to domain-specific applications. LLMs are trained on large amounts of data and learn complex patterns and relationships within the data. They utilize DL architectures, such as transformers, to process and generate human-like text. Examples of prominent LLMs include Generative Pre-trained Transformer 3 (GPT-3) [213] developed by OpenAI and Bidirectional Encoder Representations from Transformers (BERT) [214] introduced by Google. These models demonstrate the remarkable capabilities of LLMs in understanding and generating natural language content. LLMs are highly sophisticated language models distinguished by their extensive parameter sizes and exceptional learning capabilities. For many LLM models, like GPT-3, the self-attention module within the Transformer architecture serves as a foundational component for various language modeling operations. Transformers have significantly enhanced the landscape of NLP by efficiently managing sequential data, facilitating parallelization, and capturing complex long-range dependencies within text. LLMs have been used in NLP for various purposes and demonstrated extraordinary performance, including language translation, text generation, question answering, etc. [215] Furthermore, LLMs have made their way to various application domains, including IoT [216].

In Industrial AIoT applications, LLMs have not been explored to their full potential yet. However, a few existing studies have leveraged LLMs in Industrial AIoT. For instance, Ferrag et al. [210] introduced SecurityBERT, a BERT-based model tailored for IoT and IIoT cyber attack detection. BERT operates on the principle of the transformer architecture, focusing on understanding the context of words in a sentence from both directions (left and right of each word). Unlike traditional models that read text in a single direction, BERT analyzes the entire sequence of words at once, allowing it to grasp the full context of a word based on its surrounding words. Using this capability, BERT can understand language

TABLE 4. Summary of DL techniques for industrial AIoT.

Technique	Description	Main Contribution	Limitations
Autoencoder (AE)	An unsupervised neural network with an encoder and decoder. The encoder compresses input data to generate a code. A hidden layer connects the encoder and decoder.	• Cyber attack detection [119] • Threat detection [120] • Data handling [122]	• Less efficient for data augmentation • Increased model complexity
Convolutional Neural Network (CNN)	Efficient CV technique with convolution, pooling, and fully connected layers. Suitable for various industrial applications.	• Process automation [132] • Anomaly and cyber attack detection [135] • Recommendation systems [139] • Disease classification in healthcare [151]	• Higher computational complexity
Recurrent Neural Network (RNN)	Suitable for time-series data in PdM and fault diagnosis.	• Malware Threat Detection [159] • Identification of malicious patterns [158] • Fault diagnosis [162]	• Difficult training • Slow computations
Restricted Boltzman Machine (RBM)	RBM consists of input and hidden layers for feature extraction, dimensionality reduction, classification, regression, and feature learning.	• Dimensionality reduction • Feature learning	• High computational cost
Generative Adversarial Network (GAN)	Address imbalanced datasets and support anomaly detection.	• Imbalance dataset problem [176] • Data synthesis [173] • Anomaly detection [182]	• Instability and non-convergence
Deep Reinforcement Learning (DRL)	Combines ANN and RL for problem-solving.	• Computational task offloading [195] • Resource management [197]	• High computational resources required
Transformer	Encoder-decoder architecture for time-series forecasting.	• Time-series prediction [209] • Traffic analysis [204] • RUL estimation [207]	• High computational and memory requirements
Large Language Models (LLMs)	Leverage extensive datasets for content understanding and generation.	• Cybersecurity [210] • Voice-interaction systems [211] • Data augmentation in cybersecurity applications [212]	• High computational and memory requirements • Extensive parameter size • Training requires large datasets

features more deeply, making it highly effective in tasks like sentiment analysis, question answering, and language inference. SecurityBERT embeds a unique methodology centered around Privacy-Preserving Fixed-Length Encoding (PPFLE) and a byte-level Byte-Pair Encoding (BBPE) tokenizer. This approach begins with extracting and structuring network

traffic data into a BERT-comprehensible format, employing PPFLE for transforming data into text while ensuring privacy through hashing. The BBPE tokenizer then prepares this data for SecurityBERT processing. Through the extensive training and fine-tuning process that includes text normalization and the addition of special tokens, the model is optimized for

detecting a wide range of cyber threats. This methodology not only highlights the potential of applying advanced NLP techniques in cybersecurity but also demonstrates exceptional accuracy in threat classification, setting a new standard in the domain.

Furthermore, Yu et al. [217] explored the use of LLMs in Industrial AIoT for enhancing cybersecurity, particularly in the context of Advanced Persistent Threats (APTs). APTs are sophisticated and targeted cyberattacks characterized by stealthiness, persistence, and continuous efforts to breach security defenses [218]. APTs often involve well-resourced attackers who aim to maintain unauthorized access to a network over an extended period to steal sensitive information or disrupt operations. The authors present an innovative approach that utilizes BERT to analyze and identify complex, multi-stage APTs. This methodology not only significantly advances the accuracy and efficiency of threat detection systems in critical infrastructure but also demonstrates the transformative potential of integrating LLMs with cybersecurity frameworks to counter sophisticated cyber threats effectively.

Voice-interaction-based control systems in industrial settings employ voice commands to manage and control industrial devices. It enables hands-free operation while enhancing efficiency and safety across various sectors. Semantic sorting is crucial in these systems for organizing and prioritizing voice commands. It ensures more accurate and contextually relevant responses in Industrial AIoT applications. In this context, Wang et al. [211] presented a framework for deep semantics sorting in voice-interaction-enabled ICSs. It employs advanced models including multilayer CNNs and BERT. Specifically, the authors have used BERT to analyze and understand the semantics of voice commands leveraging its natural language processing capabilities to interpret the intent and context of commands accurately. This approach enhances the efficiency and accuracy of voice interactions within industrial settings. It facilitates a deeper understanding and processing of semantic relationships, thereby optimizing question-answering capabilities essential for IIoT environments. This innovative application of voice-interaction-based control systems illustrates the transformative potential of integrating DL and NLP technologies, such as BERT, in industrial operations for smarter, hands-free communication methodologies and more intuitive interactions between operators and machines.

In [212], the authors discussed the use of GPT for enhancing data augmentation in the context of multiclass intrusion detection within Industrial AIoT networks. The research primarily focused on leveraging GPT to generate synthetic network traffic data. It aims to address the challenges posed by imbalanced datasets in Industrial AIoT cybersecurity applications. By employing GPT alongside traditional interpolation-based data augmentation methods like Synthetic Minority Over-sampling Technique (SMOTE), the authors aimed to enrich the dataset to ensure a more balanced representation of various classes of network traffic.

This augmentation was intended to improve the performance of ML models for detecting a wide array of cyber threats. The experimental results demonstrate GPT's capability to synthetically generate data that closely mimics authentic network traffic patterns. Through this innovative approach, GPT is proved effective in significantly enhancing the dataset's quality and diversity, leading to more accurate and reliable cyber threat detection in Industrial AIoT environments.

To summarize, LLMs such as BERT and GPT are pioneering new frontiers in Industrial AIoT, from enhancing cybersecurity to enabling sophisticated voice-interaction systems and advancing threat detection through improved data augmentation. Their capabilities offer precise threat classification, adept voice command interpretation, and the generation of well-balanced datasets. Beyond these applications, the potential of LLMs in various industrial AIoT applications such as RUL prediction, anomaly detection, recommendation systems, etc., should be explored for optimizing operational efficiency, predicting maintenance needs, and avoiding costly downtimes. Further exploration of LLMs is needed for quality control and supply chain optimization, promising a future of enhanced automation, reliability, and insight across industrial operations. This convergence between LLM capabilities and Industrial AIoT needs underscores a transformative shift towards more secure, efficient, and intuitive industrial environments, setting a new standard for industrial technology integration.

C. COMPARISON OF ML AND DL TECHNIQUES IN INDUSTRIAL AIOT

This section discusses the comparison of ML and DL techniques in Industrial AIoT. We have discussed a range of AI techniques being vastly used in industrial settings. The exploration of ML and DL techniques such as FL and CNN, among others, highlights a transformative wave of innovation across industrial settings. By using Industrial AIoT techniques, manufacturing processes are becoming more intelligent, data-driven, and efficient. It results in significant improvements in PdM and anomaly detection. Each technique brings a unique set of strengths and applications to the industrial sector, distinguishing them from one another and showcasing their complementary roles. For instance, ML techniques such as DT and RF are suitable for classification and regression tasks, making them suited for predicting equipment failures or optimizing production lines. In contrast, DL has the ability to process vast amounts of data through layers of neural networks and excels in complex pattern recognition tasks, such as vision tasks or speech recognition in automated quality control systems.

For instance, FL stands out for its approach to data privacy and decentralized learning. By allowing AI models to be trained directly at the data source without the need of sharing the data itself, it facilitates the development of personalized AI solutions in industrial settings where data security and privacy are crucial. Meanwhile, CNNs are revolutionizing areas like PdM and quality control

TABLE 5. Comparative analysis of DL and ML in industrial AIoT.

Criteria	Deep Learning (DL)	Machine Learning (ML)
Data Requirements	Needs vast datasets to learn effectively, due to complex model architectures.	Can learn from smaller datasets, often requires data preprocessing.
Computational Complexity	Significantly high, requires Graphics Processing Units (GPUs) for efficient processing.	Relatively low, can run on Central Processing Units (CPUs).
Interpretability	Low, models are complex “black boxes”.	Higher, especially with simpler models that are easier to interpret.
Model Training Time	Long, due to complex models and large datasets.	Shorter, simpler models and less data lead to quicker training.
Use Cases	Image and speech recognition, complex pattern detection in Industrial AIoT.	PdM, anomaly detection, straightforward pattern recognition.
Scalability	Highly scalable but requires substantial computational resources.	Scalable to a degree, with less computational demand.
Accuracy	Extremely high, given sufficient data and computing power.	Varies, generally lower on complex tasks without feature engineering.
Preferred Scenarios	Scenarios that involve industrial time-series data, high-dimensional data, complex pattern recognition requiring automated learning.	Scenarios with limited data or computational resources, where model interpretability is key.

through their superior image processing capabilities. The distinct nature of these AI techniques in terms of data handling, computational complexity, and application areas implies a sophisticated approach to their integration into industrial settings. The choice of technique often depends on the specific challenges at hand, the nature of the data available, and the computational resources available. For instance, while CNNs may be preferred for tasks requiring image analysis, simpler ML models might be more cost-effective for less complex predictive tasks with structured data. Transformers and their variants are more suitable for long sequence time-series problems. However, the integration of AI into industrial operations is not without challenges. Issues such as data quality, model interpretability, and the need for substantial computational resources can limit the deployment of AI solutions. All the Industrial AIoT-related challenges are discussed further in Section V. The continuous advancements in AI technologies and the growing recognition of these challenges are paving the way for more robust, efficient, and scalable AI applications in the industrial domain. The AI techniques, each with their distinct functionalities, are intrinsically different from one another, which makes them suitable for varied scenarios. A detailed comparison underscoring these differences and their suitability for diverse scenarios is provided in Table 5.

The comparative analysis presented in Table 5 discusses the distinct characteristics and applicable domains of DL and ML within Industrial AIoT. DL excels especially in tasks when there are large datasets and powerful computers. Conversely, ML is adept at learning from smaller datasets and requires less processing power, making it suitable for tasks like simple anomaly detection. The more transparent decision-making process of ML presents a contrast to DL's reduced interpretability as a ‘black box’ model. Despite these variations, both strategies are immensely beneficial to the industrial sector, their application primarily depending on the

type of data involved, the available computational power, and the specific needs of the task at hand.

IV. KEY APPLICATIONS OF INDUSTRIAL AIOT

In this section, we explore the potential key applications of Industrial AIoT. Industrial machines and devices, equipped with numerous sensors, generate a substantial amount of data. The intelligent processing of this data is crucial to extract meaningful insights and valuable information to improve the decision-making process. The applications of Industrial AIoT are diverse, as shown in Fig. 8. The architecture of Industrial AIoT, illustrated in Fig. 1, demonstrates that while the core principles of Industrial AIoT, such as enhancing efficiency, data analysis, and automation, are generally consistent across industries, their application and impact can differ significantly. This leads to unique principles tailored to the specific needs and challenges of each industrial application. For instance, in the manufacturing industry, Industrial AIoT principles often focus on PdM, real-time equipment monitoring, and the optimization of production lines for efficiency. Here, the unique principle involves integrating AI to predict machine failures and proactively schedule maintenance, thus reducing downtime. In healthcare, Industrial AIoT principles emphasize patient monitoring, personalized medicine, and data analysis for improved diagnostics and treatment planning. The critical unique aspect here is ensuring patient privacy and data security while utilizing AIoT for proactive health management. Similarly, in agriculture, Industrial AIoT applications are directed towards precision farming, crop health monitoring, and task automation, such as irrigation. The unique principle in agriculture is employing AI to parse data from IoT devices for real-time decision-making that enhances crop yield and minimizes waste. Hence, in all industrial applications, the core principles of Industrial AIoT remain consistent. However, the application and impact are requirement-driven for every industry and vary accordingly.

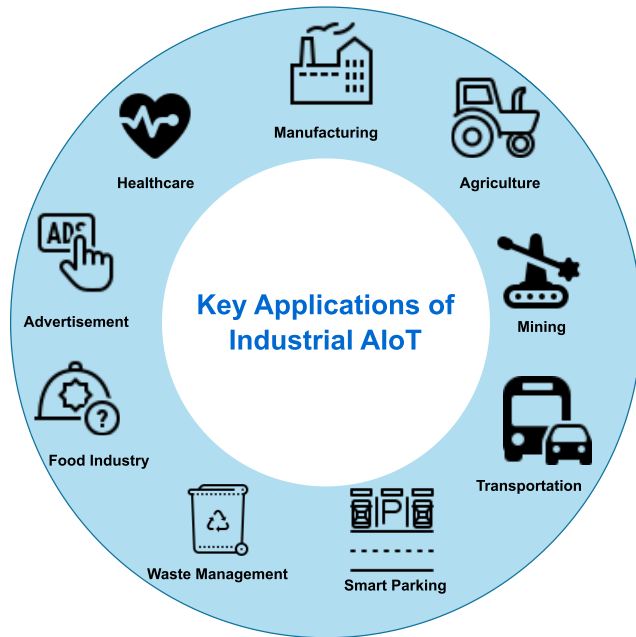


FIGURE 8. Key applications of industrial AIoT.

A. INDUSTRIAL AIOT IN THE REAL-TIME MANUFACTURING INDUSTRY

Industrial AIoT techniques are remarkably transforming the manufacturing industry. Industrial sensors enable the monitoring of machine performance in real time, helping to avoid unnecessary production delays. Real-time manufacturing data analysis is critical for the safety of the product and quality control operations. Additionally, this real-time analysis assists in PdM of machines, thereby reducing the failure rate. In the manufacturing industry, Industrial AIoT techniques are employed to process and analyze manufacturing data. This enables the detection of anomalies in production processes, facilitates timely disruption management, automates manufacturing processes, and increases overall production efficiency [219]. For instance, consider the scenario of a large-scale automotive manufacturing plant that leverages Industrial AIoT techniques. By integrating sensors across the production line, the plant continuously monitors machine performance and detects anomalies in real time. This enables proactive maintenance interventions, minimizing downtime and optimizing production efficiency. Similarly, for the intelligent manufacturing, a case study of Siemens industry is presented in [220]. The integration of technologies such as AI, IIoT, Digital Twins, and Cloud Computing has driven Siemens' manufacturing revolution. Siemens is a worldwide technology and engineering powerhouse which has applied these technologies to shift away from traditional manufacturing practices and toward an intelligent manufacturing model. This revolution is marked by the automation and digitization of industrial processes, with smart sensors and IoT devices playing critical roles in data gathering and analysis, enabling PdM and efficiency opti-

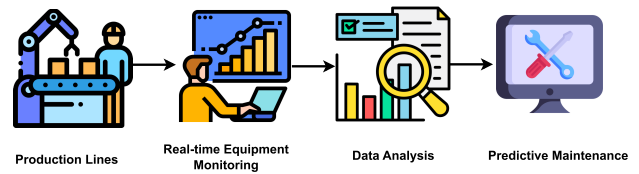


FIGURE 9. Industrial AIoT enabled real-time manufacturing industry.

mization. ML algorithms analyze massive volumes of data to forecast equipment failures before they occur, resulting in significant downtime and maintenance savings. Digital Twins generate virtual versions of real systems to mimic, monitor, and manage the manufacturing process, improving product quality and operating efficiency. Cloud computing provides scalable and flexible data storage and analysis, promoting innovation and collaboration across global manufacturing facilities. Siemens' example demonstrates how the integration of sophisticated technologies may transform manufacturing, resulting in higher productivity, cost savings, and a competitive advantage in the digital era.

Usually in the manufacturing sector sensors, robots, smart tools, and historical data are used. The AI models process the data to perform functions such as identifying and tracking humans, recognizing assembly parts, assessing human activities, and understanding the assembly environment. The outcome of this processing then aids human operators, enabling them to work more efficiently. Through automated analysis of manufacturing data, the plant can detect and address production disruptions promptly, ensuring smooth operations. This example illustrates how Industrial AIoT techniques drive operational excellence and efficiency in manufacturing environments, showcasing the practical applications of real-time monitoring and analysis in enhancing production processes.

In real-time manufacturing, Industrial AIoT principles often focus on PdM, real-time monitoring of equipment in the production lines for effective data analysis, and enhanced efficiency. Fig. 9 shows the activities in the Industrial AIoT-enabled real-time manufacturing industry. However, Industrial AIoT techniques require significant amounts of data to make accurate decisions about fault identification, defect classification, PdM, and quality control [33]. To address this challenge, innovative approaches such as data synthesis and transfer learning techniques are being explored [173], [221].

B. INDUSTRIAL AIOT IN AGRICULTURE INDUSTRY

In many countries, agriculture significantly contributes to economic growth [222], [223]. Industrial AIoT is transforming this industry by making farming smarter and more efficient [224], [225], [226], [227], [228]. It replaces traditional manual techniques with smart farming methods, thereby enhancing crop production. Industrial AIoT in agriculture effectively handles a variety of challenges, providing innovative solutions. For example, AI-assisted drones are used for pest control, crop monitoring, identifying diseased

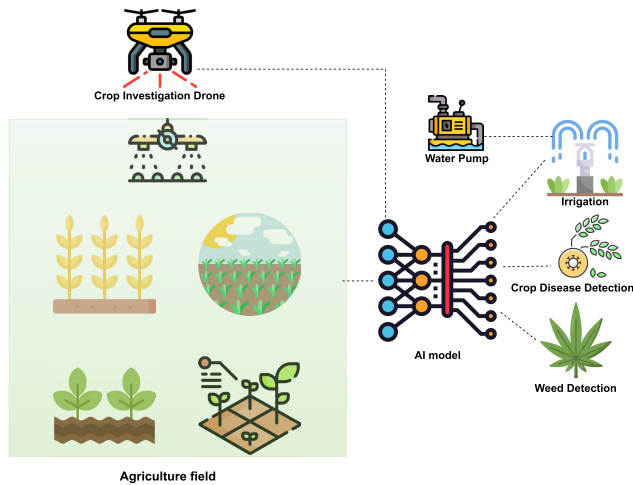


FIGURE 10. Industrial AIoT enabled agriculture industry.

crops, and targeted spraying [229]. AI also plays a crucial role in food monitoring, quality control, and managing soil and irrigation systems [229]. For example, Industrial AIoT transforms irrigation systems by integrating smart sensors and AI algorithms [230]. These systems continuously monitor soil moisture levels, weather conditions, and crop water requirements. Based on real-time data and predictive analytics, Industrial AIoT enabled irrigation systems autonomously adjust watering schedules to optimize water usage and ensure crop health. This intelligent irrigation approach minimizes water waste, enhances crop yields, and promotes sustainable agricultural practices, showcasing the multifaceted benefits of Industrial AIoT in modern farming.

Image-based AI techniques, such as CNNs, are increasingly popular for plant disease detection in agriculture [231], [232]. This process involves capturing leaf images and using AI for feature extraction to identify patterns indicative of disease, thereby facilitating accurate disease classification. Another significant application of Industrial AIoT is in weed identification [233], [234], where CNN models trained on weed images can effectively differentiate between weeds and crops. Additionally, soil sensors measure moisture levels, and the data is processed to identify areas requiring irrigation, enabling precise water management. This intelligent integration of technology exemplifies the advancements of Industrial AIoT in the agriculture industry. Fig. 10 shows the Industrial AIoT enabled agriculture industry.

C. INDUSTRIAL AIOT IN MINING INDUSTRY

Mining is one of most prominent economic industries that has evolved over time and demands real-time data analysis due to its nature of operations. Minerals are essential for humans and we require them daily. Over time, the increased demand for minerals has resulted in the expansion of the mining industry. According to the IBM report [235], each individual needs about 3.11 million pounds of fuel, metals, and minerals in their lifetime. The mining industry

involves costly and risky operations [236]. Industrial AIoT has the potential to increase mining production while reducing risks, waste, and unnecessary costs involved in its operations.

Besides, the mining industry involves numerous types of risks. For instance, in the coal mining industry, chemical reactions, hazardous gas emissions, rock sliding, and suffocation are potential risks that can be dangerous for the mining workforce. Therefore, heterogeneous sensors can detect chemical reactions, gas emissions, and unsafe events data [183]. The collected data can be analyzed using AI techniques to detect risky events and inform the related personnel in time.

Furthermore, sensors can also be used to collect data before the digging process begins, allowing effective decisions to be made ahead of time [237]. Additionally, AI techniques can also bring intelligence and automation to the mining industry to make the mining operations autonomous where no human input is required [238], [239]. For instance, in a large-scale underground mine, Industrial AIoT systems can integrate a network of sensors to monitor various parameters such as temperature, gas levels, and structural integrity in real-time. These Industrial AIoT techniques analyze this data to predict potential hazards like gas leaks or structural instability. It allows for timely intervention to prevent accidents. Furthermore, autonomous drones equipped with AI-powered cameras can survey remote or hazardous areas, providing real-time visual data to aid in decision-making and ensuring worker safety. Such applications of Industrial AIoT can revolutionize safety and efficiency in the mining industry, making operations more sustainable and productive while safeguarding human lives.

Furthermore, the digital twin framework can also revolutionize mining safety practices when combined with industrial AIoT [240]. Similar to its role in predicting dangerous events, the system can predict fire hazards by monitoring parameters such as temperature and gas levels. Furthermore, drones equipped with AI enabled cameras can also be used to survey underground tunnels to provide real-time visual data for fire detection and prompt evacuation as presented in [240]. Consequently, safety and operational efficiency are enhanced, thereby ensuring the well-being of mining personnel and ensuring sustainable operations. The unique underlying principle governing the implementation of Industrial AIoT in the mining industry is the prioritization of safety through advanced hazard detection and predictive analytics. By deploying a network of sensors and AI-driven technologies, mining operations are not only made safer by predicting and preventing potential accidents but also more efficient, allowing for real-time monitoring and autonomous operations that minimize the need for human exposure to dangerous conditions. This enables predictive and autonomous operations that significantly enhance both safety and efficiency, showcasing a unique approach tailored to meet the specific challenges of the mining sector. The unique and distinct principle of employing Industrial

AIoT in the mining sector lies in the prioritization of safety through innovative hazard detection and predictive analytics. This approach facilitates not only a safer working environment by foreseeing and mitigating potential risks but also optimizes operational efficiency through real-time monitoring and autonomous processes that reduce human exposure to hazardous conditions.

D. INDUSTRIAL AIOT IN TRANSPORTATION INDUSTRY

The transportation industry plays a crucial role in the economy of any country. It encompasses commercial buses, trains, cargo vehicles, public transportation, and private automobiles. Autonomous vehicles have gained significant attention and are becoming increasingly popular [241], [242]. Furthermore, Industrial AIoT-enabled intelligent transportation systems (ITSs) include roadside units (RSUs). These RSUs are equipped with intelligent models and techniques that enable the Internet of Vehicles (IoV) to provide a variety of services, including entertainment, context-aware, and location-based services.

Additionally, Industrial AIoT applications in the transportation industry encompass advancements in intelligent autonomous vehicles, managing transport flow, smart traffic lights, and travel time prediction. For example, Tesla employs deep learning models to enhance the capabilities of self-driving cars and ensure safety [243]. Fig. 11 illustrates Industrial AIoT in the Transportation Industry. The integration of Industrial AIoT into the transportation sector is exemplified by the development of the MetroPT dataset for PdM in public transport vehicles [244]. The authors of the dataset installed a signal acquisition system in the Air Production Unit (APU) of a train for collecting data from analog sensors, digital signals, and GPS information for a metro public transportation service in Porto, Portugal. The collected data encompasses variables such as pressure on the compressor, oil temperature, motor current, and more. This data was collected using sensors (Industrial AIoT devices) to facilitate online anomaly detection and failure prediction, shifting the maintenance paradigm from reactive to predictive. The ultimate goal is to predict developing and unexpected failures based on continuously monitored equipment conditions for minimizing unplanned downtime and associated costs. This approach not only enhances the efficiency of transportation services but also significantly contributes to their reliability and safety. It illustrates the transformative potential of Industrial AIoT in the transportation industry. In this way, using Industrial AIoT technologies, the reliability of public transportation can be enhanced, and downtime can be reduced. The distinct underlying principles of Industrial AIoT implementations in the transportation industry emphasize enhancing system inter-connectivity and real-time data analysis to improve service reliability and operational efficiency, underscoring the shift towards more predictive and proactive maintenance strategies for safer and more efficient autonomous travel.

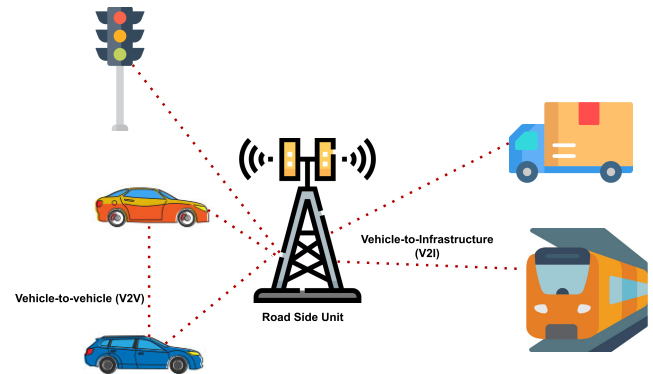


FIGURE 11. Industrial AIoT enabled transportation.

E. INDUSTRIAL AIOT IN SMART PARKING

People migrate to cities for better job opportunities, health facilities, and higher education, leading to densely populated urban areas [184]. The growth of the transportation industry and the increase in vehicle usage as primary commuting resources have exacerbated the challenge of finding vacant parking spaces in major cities like Toronto, New York, and Los Angeles. A report highlights the staggering cost of searching for parking in Los Angeles, which amounts to 730 tonnes of CO₂, 95,000 hours, and 47,000 gallons of gasoline annually [245]. Industrial AIoT-based solutions offer effective strategies to mitigate these parking issues [246], [247], employing AI techniques for detecting vehicle presence and counting vacant slots, with image-based AI models like CNNs playing a pivotal role [248], [249]. Additionally, predicting and forecasting parking occupancy with methods such as the DRL approach proposed by Awaisi et al. [190] can enhance parking space management during peak hours. AI techniques are crucial not only for predicting parking occupancy or availability [250] but also for calculating variable parking prices based on observed parking slot occupancy [251].

The Smart Parking System implemented by San Francisco in 2011, known as “SFpark,” serves as a practical example of Industrial AIoT’s impact on urban parking [251]. According to [252], SFpark introduced a demand-responsive pricing system that dynamically adjusted meter rates based on real-time parking space occupancy, covering 6000 metered spaces in seven districts or 25% of the city’s on-street parking. This program aimed for an optimal occupancy rate of 60–80% for each block face. Moving away from the static pricing model that varied only by location, SFpark’s variable pricing increased or decreased within specific time bands—morning, early afternoon, evening, and night—depending on occupancy rates. Rates would rise by \$0.25 per hour if occupancy exceeded 80% and decrease by \$0.25 to \$0.50 if it fell below 60%, with the goal of encouraging more efficient usage. Rate changes were announced on SFpark’s website

at least seven days in advance, with a maximum rate of \$8.00 and a minimum of \$0.25 per hour. Non-price changes, such as enabling credit card payments and extending parking durations, were also implemented to make parking more user-friendly. Through strategic adaptations, SFpark demonstrated how technology and data analytics could revolutionize urban parking management, significantly enhancing efficiency and convenience, and setting a global benchmark for smart parking solutions.

In conclusion, Industrial AIoT implementations in smart parking systems like SFpark highlight a transformative approach to urban parking management, leveraging real-time data analysis and AI technologies to optimize parking space utilization and reduce urban congestion. The distinct underlying principles of these implementations center around demand-responsive pricing, predictive occupancy modeling, reducing CO₂ and time for parking search, and enhanced user convenience. By integrating such principles, smart parking solutions not only address the immediate challenges of parking in densely populated areas but also lay the foundation for more sustainable and efficient urban transportation ecosystems.

F. INDUSTRIAL AIOT IN WASTE MANAGEMENT INDUSTRY

The exponential growth of population and rapid urbanization are among the potential causes of increased waste generation worldwide. According to the World Bank report, 2.01 billion tonnes of solid waste were produced in 2016, a number projected to rise to 3.04 billion tonnes by 2050 [253]. Consequently, making the waste management process efficient and smart is a key objective in the industry [254], [255]. Smart waste management systems can prevent the spread of various diseases. The overall process includes waste collection, disposal, and recycling. AI techniques are highly effective in addressing waste management challenges, such as predicting and classifying waste generation [256], [257], [258], [259], [260], [261]. In the waste management industry, AI techniques can speed up the recycling process and ensure proper waste disposal by managing garbage collection points. Image-based AI techniques are instrumental in classifying waste into categories like glass, garbage, organic, paper, and plastic. Notably, the most extensively used AI techniques in this sector include ANN, SVM, linear regression (LR), DT, and GA [262].

Further enhancing the efficiency of waste management systems, Industrial AIoT can provide real-time monitoring and management solutions. It can facilitate a more dynamic and responsive approach to waste management, enabling systems to not only predict waste generation but also to actively monitor fill levels of bins and optimize collection routes based on actual data. This integration represents a significant leap towards creating sustainable urban environments that reduces operational costs, and minimizing environmental impacts. Through smart sensors and IoT devices, waste management becomes a highly efficient, data-driven process,

offering insights into waste patterns and enabling cities to adapt their strategies accordingly.

Building on these advancements, the authors in the study [263] discussed the practical application of these technologies at Ton Duc Thang University, Vietnam, illustrating the practical benefits of integrating Industrial AIoT into waste management. By employing sensors to monitor the fill levels of trash bins and using ML algorithms specifically, LR to predict waste accumulation, this system significantly enhances the efficiency of waste collection. It optimizes collection routes and schedules, leveraging graph theory to determine the most efficient collection paths. This case study exemplifies how the Industrial AIoT technologies can not only address traditional waste management challenges but also contribute to more sustainable and efficient urban ecosystems, aligning closely with the broader objectives of the industry to adapt and innovate in response to growing environmental concerns. The core principles governing Industrial AIoT implementations in waste management revolve around real-time data analysis for informed decision-making, predictive analytics to optimize operational efficiency, and sustainability through the reduction of environmental impacts. These foundations enable dynamic adjustments to waste collection processes, ensuring they are both efficient and adaptive to changing urban landscapes, ultimately contributing to smarter, more sustainable waste management practices.

G. INDUSTRIAL AIOT IN THE FOOD INDUSTRY

The food industry is among the fastest-growing industries in the world, encompassing operations such as food packaging, frozen food, and organic waste analysis [15]. The Industrial AIoT use in the food industry which has been increasing in recent years, supports a range of functions including food sorting, classification, quality control, safety, supply chain management, production process development, and maintaining industrial hygiene [264]. With the global population expected to rise, food demand is anticipated to increase by 59% to 98% by 2050 [265]. Thus, industrial AIoT in the food industry will play a crucial role in meeting future food demands. Beyond improving packaging, quality control, supply chain management, and organic waste management, the impact of Industrial AIoT will also extend to enhancing business strategies, notably through the capabilities of AI in sales prediction and food yield optimization.

One prominent use case of Industrial AIoT in the food industry is supply chain management. Food waste is a worldwide issue and can be resolved with effective supply chain management. The United Nations has highlighted a significant issue in global food distribution, noting that annually, approximately one-third of all food produced equal to a value of \$750 billion and cultivated on 28% of the world's agricultural land is wasted [266]. According to [266], the global market for AI applications in supply chains was valued at \$491 million in 2017 and is predicted to reach \$6,548

million by 2024. It shows how AI and IoT in revolutionizing supply chain operations. The integration of Industrial AIoT in the food industry offers potential solutions to global challenges. It can provide real-time monitoring and decision-making in the food industry by using IoT devices equipped with sensors, and AI. For example, IoT technologies enable the tracking of environmental conditions that can affect food products in transit. It facilitates immediate adjustments to ensure freshness and food safety. Concurrently, AI and big data play crucial roles in optimizing logistics, predicting demand, and managing inventory. It can significantly reduce waste and enhance the efficiency of supply chains. This collective application of Industrial AIoT technologies not only addresses the supply-demand fluctuations inherent in the food industry but also meets high safety and sustainability objectives. The fundamental principles guiding Industrial AIoT in the food industry center on precision agriculture and predictive supply chain management, harnessing real-time data and AI analytics to optimize food production, distribution, and reduce waste, thereby ensuring food security and sustainability in an ever-growing global demand scenario.

H. INDUSTRIAL AIOT IN THE ADVERTISEMENT INDUSTRY

Advertisement plays a crucial role in introducing new products to the market, expanding businesses, and boosting sales. In the past decade, the use of social media platforms such as Facebook, YouTube, Twitter, and Instagram has increased dramatically. Consequently, traditional advertisement methods have become less effective. A key strategy in modern advertising is targeting the right audience for services or products. To this end, the advertisement industry is increasingly adopting AI to accurately target audiences on social media platforms. Social media platforms employ AI-based recommendation algorithms for effectively advertising products and services [267]. Industrial AIoT techniques are key not only for increasing sales and business expansion but also for analyzing local trends to tailor billboard advertisements more effectively. These techniques are also actively used to display advertisements tailored to users' previous activity data, thereby enhancing the relevance and impact of advertising campaigns.

Industrial AIoT has the potential to transform the advertising industry by allowing for better, more focused advertising techniques that use real-time data to make decisions. Businesses that integrate IoT devices with AI analytics may collect and analyze massive volumes of data on customer behavior, preferences, and environmental factors. This enables dynamic modification of adverts, ensuring they reach the most appropriate audience at the best moment. For example, digital billboards may alter their displayed advertising according on traffic flow, weather, and social media trends, whilst internet platforms can provide individualized ad experiences based on user behavior. This combination of IoT connection and AI predictive skills not only improves the success of advertising campaigns, but

also opens up new possibilities for connecting with customers, making advertising more interactive and responsive. In the advertisement industry, Industrial AIoT is distinct by the principles of real-time data analysis and targeted consumer engagement. These foundational elements enable the creation of highly personalized and contextually relevant advertisements, optimizing both the impact and efficiency of advertising campaigns.

I. INDUSTRIAL AIOT IN HEALTHCARE INDUSTRY

Healthcare is a key industry offering valuable health services globally. Over time, it has evolved rapidly to meet the medical needs of people [268] and has become one of the top revenue-generating sectors in many countries. The healthcare industry has shown its critical importance in times of crisis, notably during the COVID-19 pandemic, where it played a pivotal role in providing essential health-related services [269]. Technological advancements play a significant role in the healthcare industry, including smart billing, patient-record management, and monitoring [270].

The advent and progress of digital data acquisition techniques have enabled AI to expand into areas traditionally dominated by human expertise. The applications of AI in healthcare are diverse, ranging from cancer and tumor detection to medical imaging diagnosis [271]. AI-enabled healthcare applications utilize existing medical data and patients' health records, aiding doctors in making efficient diagnoses and providing valuable treatments [272]. Hospitals have invested billions of dollars in AI, with an estimate suggesting that AI in healthcare could generate an annual profit of \$150 billion for the U.S. healthcare industry by 2026 [273]. A critical application of Industrial AIoT in healthcare is the early diagnosis of diseases, such as cancer, which are challenging to detect in their initial stages [274], [275], [276], [277], [278], [279]. AI classification techniques, trained on supervised learning models, are proving useful in disease diagnosis. Medical imaging diagnosis, for instance, benefits from AI-driven Computer Vision techniques [280], [281]. Furthermore, AI in healthcare has promising applications in health-record management, drug discovery, and disease outbreak prediction. Fig. 12 illustrates the AI applications in healthcare industry.

Among prominent uses and case studies of Industrial AIoT in healthcare system, one is the use of Industrial AIoT for voice pathology detection. The voice pathology detection systems utilize advanced audio processing techniques and ML algorithms to analyze voice signals for early detection of diseases. The authors in [282] introduced an IoT-based ML system for diagnosing voice pathologies. By utilizing microphones and electroglottography devices for signal capture, and processing these through pretrained CNN and LSTM networks. This approach achieves a remarkable 95.65% accuracy in voice pathology detection. This case study underscores the integration of Industrial AIoT in creating efficient, predictive models for early

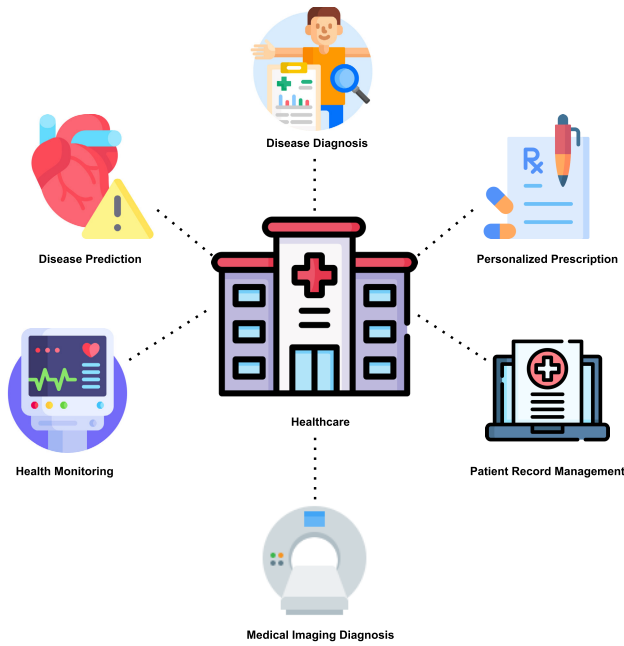


FIGURE 12. Industrial AIoT enabled healthcare.

disease diagnosis, aligning with the healthcare industry's goals of adopting innovative solutions to meet increasing medical demands. The distinct underlying principles in the implementation of Industrial AIoT in healthcare emphasize real-time data acquisition and analysis, predictive analytics for early diagnosis, and the integration of devices and systems for comprehensive health monitoring. It also focuses on adaptability, and comprehensive monitoring for ensuring timely and accurate diagnoses while advancing towards more sustainable and efficient healthcare ecosystems.

V. KEY CHALLENGES IN INDUSTRIAL AIOT

As an emerging technology, Industrial AIoT is associated with a series of technical challenges. In this section, we describe the key challenges involved in Industrial AIoT applications. Fig. 13 includes the challenges that will be discussed as follows.

A. DATA AVAILABILITY

AI-assisted industrial automation is heavily dependent on the availability of high-quality, relevant data for effective implementation. The absence of such data presents a significant challenge, particularly affecting the efficiency and accuracy of AI techniques [173]. Supervised learning algorithms, commonly used in Industrial AIoT applications, require a substantial amount of data to avoid under-fitting. An under-fitted model can lead to prediction errors, elevated loss metrics, and compromised prediction accuracy [173]. This issue becomes particularly noticeable in industrial systems that are either newly implemented or have not been operational for an adequate duration. In such cases, the

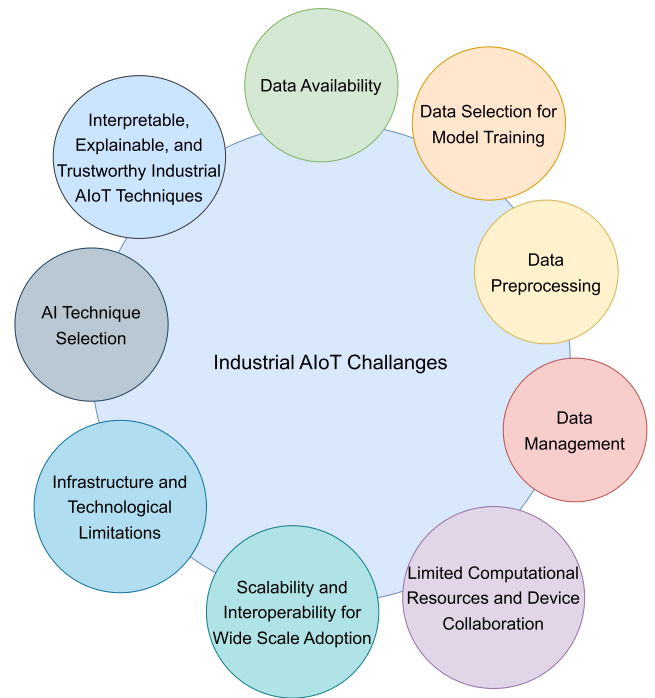


FIGURE 13. Industrial AIoT challenges.

absence of a robust historical dataset can severely limit the training of the model, affecting its performance.

Another significant issue in Industrial AIoT pertains to imbalanced datasets—characterized by an abundance of operational ('normal') state data in contrast to the limited availability of failure ('faulty') state data. Since mechanical components are often replaced or refurbished before failure, this leads to a dataset heavily skewed towards the normal state, posing a significant challenge for model training and the accurate prediction of faults [283]. The predominance of normal state data can lead AI techniques to become biased towards predicting normal states, reducing their sensitivity to anomalous or failure conditions. This bias can significantly impact the reliability of predictive systems, which are critical for ensuring operational efficiency.

In some cases, real-world data may not be available for specific industrial applications, such as PdM of turbofan engines. In these cases, simulated data can serve as a valuable technique for generating datasets that mimic real-world conditions. For example, the C-MAPSS tool has been developed within the Matlab/Simulink environment by Mathworks, Inc. to simulate the behavior of large commercial turbofan engines [284]. The C-MAPSS dataset emulates real-world turbofan engine conditions and provides real-time degradation information. Introduced in [285], this dataset has been extensively used and verified by numerous researchers [207], [286], [287], [288], [289]. Consequently, such simulated datasets have become benchmarks for model training. However, when selecting a simulated dataset, it is crucial to ensure that it closely approximates real-world

conditions, covers a comprehensive range of scenarios, including both healthy and faulty data, and has been validated by the research community. Despite these criteria, finding a simulated dataset that fulfills all these requirements remains a challenge.

In summary, the success of AI in industrial automation heavily relies on the availability of high-quality, diverse data. Key challenges include the scarcity of historical data, leading to under-fitted models, and the prevalence of imbalanced datasets that are biased towards normal operational states. This bias affects the accuracy of fault detection. Simulated datasets, such as the C-MAPSS tool for turbofan engines, are valuable but finding ones that accurately represent real-world conditions and are validated by the research community remains a challenge.

B. DATA SELECTION FOR MODEL TRAINING

Choosing the right data for model training is a crucial step. The success of Industrial AIoT techniques in solving specific industrial problems depends on selecting the correct training dataset. Therefore, it is essential to choose training data that avoids biases and is well-suited for addressing the particular industrial problem. For example, PdM data is not suitable for training if it does not include all possible equipment degradation scenarios. In such cases, achieving optimal model accuracy and performance becomes impossible. Furthermore, the data should be diverse enough to capture various aspects of the problem, including different operating conditions, machine states, and other variables that could affect the performance of the model. For instance, in the case of anomaly detection, the data should include a variety of real-world anomalies for effective model training.

The task of selecting appropriate data for model training is not merely a preliminary step but a significant challenge in itself. The choice of data can make the Industrial AIoT model effective in solving industrial problems. Therefore, careful consideration and planning are required in the data selection process to ensure the success of the Industrial AIoT model.

C. DATA PREPROCESSING

Once data is selected and available, the subsequent crucial step is its preprocessing. Data must undergo preprocessing to be suitable for training AI models. Often, raw data is not in a form ideal for these models, containing missing values, irrelevant or erroneous entries, and outliers. These issues render the data unsuitable for effective model training. Preprocessing activities encompass steps such as converting data into appropriate formats, cleaning data, and addressing missing values. In the context of Industrial AIoT, preprocessing is essential to obtain meaningful insights and ensure effective model training. A significant challenge in this process is the absence of sensor data, often due to transmission errors or hardware failures. In such cases, one potential solution is to estimate the missing values through a process called imputation. Various methods for imputation exist [290], but

the primary challenge is to approximate the true data points closely. Imputation can unintentionally introduce bias into the dataset, leading to potentially unfair predictions. The reliability of the model depends on both the nature of the missing data and the selected imputation method [291]. It is important to highlight that every imputation method brings in biases, and research has revealed that estimated values can deviate significantly from actual values, making it a significant challenge [291].

Additionally, in the data preprocessing stage, selecting features with the highest relevance, least redundancy, and optimal characteristics is crucial [292]. The accuracy of predictions from Industrial AIoT models hinges on the number of relevant features. Industrial datasets often comprise numerous features, many of which are irrelevant to the predictions. Therefore, exploring various feature selection methods during preprocessing is vital. However, assessing and selecting relevant features from a large dataset remains a challenging task.

D. DATA MANAGEMENT

The heterogeneous Industrial AIoT devices generate a vast amount of data, presenting numerous data management challenges. These challenges are multifaceted and include security, data privacy, and integration issues. For the efficient and effective development of Industrial AIoT solutions, these challenges must be addressed. With the advent of ICSs and Industry 4.0, Industrial AIoT-enabled systems generate, process, and exchange vast amount of data that is privacy-sensitive and security-critical. Therefore, they become prime targets for security attacks [293]. Given the large-scale industrial development and growth, data security has emerged as an increasingly significant concern [294]. The inherent nature of connected devices and their communication networks makes Industrial AIoT systems particularly vulnerable to security issues. New types of attacks continuously threaten the security of Industrial AIoT systems. For instance, APTs slowly compromise networks to steal data, ransomware attacks restrict access until a ransom is paid, and zero-day exploits take advantage of yet-to-be-discovered vulnerabilities. To counter these evolving threats, innovative security solutions and constant monitoring are essential. Security measures must be implemented carefully since the data is sensitive [295]. Significant efforts have been made by researchers to improve and enhance the security of industrial environments, including ML-based techniques [296], [297], [298] as discussed in Section III. However, with the introduction of new kinds of security attacks and breaches every day, there remains a pressing need for continuously improved security mechanisms to ensure efficient data management and enhance data security. The security challenges also give rise to data privacy concerns in industrial settings. In some industries, for instance, the healthcare industry, sensors and devices are deployed to capture, examine, and record patients' data,

which is highly confidential. Thus, data privacy becomes a prominent aspect [299]. Similarly, in other industrial settings, such as the manufacturing industry, data privacy cannot be ignored, as data breaches can greatly impact the reputation of enterprises and the trust of stakeholders. The foundation of trust among users lies in the integrity of data privacy, which necessitates the adoption of strict protocols and encryption techniques to protect sensitive information from unauthorized access. Researchers have made substantial efforts to ensure privacy in industrial settings by introducing various techniques [300]. For instance, the authors in [1] introduced a user authentication scheme through identity management to preserve data privacy in the healthcare sector using the Elliptic Curve Cryptography technique. However, with the rise of security concerns day by day, data privacy still presents a significant challenge and must be explored further in the future to ensure privacy in Industrial AIoT-enabled systems.

Beyond security and privacy, there lies the challenge of data integration in the Industrial AIoT landscape. Industrial AIoT environments consist of a plethora of heterogeneous devices including temperature sensors, actuators, and RFID tags to complex machinery equipped with smart sensors. The devices generate data in different formats and structures. This diversity results in scattered data sources and types, presenting significant challenges in achieving a synchronized and unified data ecosystem [301]. To efficiently process and extract meaningful information, seamless integration of this diverse data is important. Therefore, the diverse nature of data types poses a challenge in Industrial AIoT systems for efficient data integration. New devices and technologies necessitate the development of novel methodologies and tools for integrating and interpreting the generated data. The success of Industrial AIoT systems lies in effectively managing and integrating this diverse data.

The security, privacy, and integration issues form a significant triad of challenges in data management within the Industrial AIoT. Addressing these interconnected issues is pivotal for advancing the frontier of smart industrial operations, ensuring not only the security and privacy of critical data but also its meaningful utilization across diverse operational platforms.

E. LIMITED COMPUTATIONAL RESOURCES AND DEVICE COLLABORATION

The industrial environment is composed of a complex network of heterogeneous industrial sensors and devices, ranging from simple temperature sensors to advanced robotic actuators [302]. These devices collectively form the backbone of the industrial environment, promising unprecedented levels of automation, efficiency, and intelligent decision-making [16]. However, the realization of this vision faces significant challenges.

One of the most significant challenges is the limited computational resources available on industrial devices.

Advanced AI techniques such as CNN, DNN, DRL, Transformer, and GAN require high computational resources for effective processing [303]. When deployed on resource-constrained industrial devices, these techniques can suffer from higher latency, excessive energy consumption, and increased network bandwidth usage. Although cloud computing has been employed to mitigate these issues, it introduces its own set of problems, such as high data transfer and processing latency, particularly for real-time, latency-sensitive applications [304].

In environments with limited computational resources, the importance of effective device collaboration is particularly crucial. This is because collaborative strategies can distribute computational tasks among devices, reducing the burden on any single device and enhancing overall efficiency. However, a significant challenge in these settings is ensuring seamless collaboration among heterogeneous devices [16]. Manufacturers often employ proprietary communication protocols and unique data formats, creating barriers to seamless device interaction. This lack of standardization not only complicates data integration but also introduces operational inefficiencies and security vulnerabilities, exacerbating the challenges of device collaboration. Effective collaboration, which is essential for devices with differing capabilities and design principles to communicate and work in harmony, is not easily achieved. The absence of seamless collaboration further complicates data integration and results in operational inefficiencies and security vulnerabilities. In environments constrained by limited resources, this lack of collaboration and standardization can lead to devices competing for processing power, bandwidth, and storage. This competition can create bottlenecks that hinder data processing and decision-making, potentially leading to sub-optimal outcomes or even system failures.

In summary, the full potential of Industrial AIoT in revolutionizing industrial operations is hindered by challenges in computational capacity and device collaboration. Enhanced collaboration among devices can help offset these limitations, thereby improving overall system performance and resilience.

F. SCALABILITY AND INTEROPERABILITY FOR WIDE SCALE ADOPTION

Industrial AIoT systems are evolving rapidly that makes scalability and interoperability crucial factors to investigate for the wide scale adoption and implementation. These factors significantly affect the success and functionality of these complex systems. As a result, it is critical to comprehend, consider, and handle the challenges of scalability and interoperability in Industrial AIoT systems.

In the context of Industrial AIoT, scalability refers to the system's ability to handle an increasing amount of work or its potential to expand to accommodate growth [305]. Scalability issues present a significant challenge for wide-scale adoption, as they involve integrating an increasing

number of devices, managing data volumes, and enhancing processing capabilities for analyzing vast datasets in real-time. As industries aim to incorporate AI techniques to improve operational efficiency and innovation, it is crucial to note that the demand for systems capable of dynamic scaling with minimal disruption is critical. This need highlights the necessity for robust, adaptable architectures and innovative data management techniques, as outlined in Section V-V-D, to adapt to the changing environment of industrial operations.

Furthermore, interoperability is defined as the seamless exchange of data between devices, platforms, and applications within an Industrial AIoT ecosystem [306]. The diversity of standards, protocols, and data formats across different devices and systems poses a significant challenge to achieving interoperability and device collaboration, as discussed in Section V- V-E. Environments that integrate legacy systems with newer IoT technologies require a unified communication framework to bridge these disparities. Ensuring interoperability is fundamental for the wide-scale adoption and full realization of Industrial AIoT's potential, as it enables cohesive operations, enhances data sharing and analytical capabilities, and supports more informed decision-making processes.

Addressing scalability and interoperability issues necessitates a collective effort toward standardization, the adoption of modular design principles, and the implementation of AI-driven analytics capable of adapting to various industrial contexts. Efforts toward standardization should aim at developing universal protocols and data formats to facilitate interoperability among diverse systems. Modular designs enable the incremental addition or modification of system components without significantly overhauling existing systems. Moreover, leveraging Industrial AIoT techniques for large-scale data analytics can optimize operations and enhance decision-making processes.

G. INFRASTRUCTURE AND TECHNOLOGICAL LIMITATIONS

AI is creating new opportunities for value creation in businesses, sectors, local communities, and society at large [307]. Industrial AIoT techniques are evolving day by day to address inherent limitations and problems such as data privacy concerns and interoperability challenges. However, these techniques still face infrastructure and technological limitations. Technological barriers significantly impact the adoption of Industrial AIoT systems. Despite their promise of transformative benefits for industries through enhanced efficiency, innovation, and decision-making processes, several key challenges limit their widespread implementation.

Deploying, managing, and operating Industrial AIoT systems requires comprehensive infrastructure. This encompasses essential physical and organizational structures, including network connectivity, data processing and storage, security systems, and power and energy supply. It highlights the need for robust, reliable, and scalable Industrial AIoT technologies. Scalability is crucial as the infrastructure

must expand to meet the growing demands of industrial activities. The Industrial AIoT environment, comprised of heterogeneous interconnected devices, necessitates foundational network and connectivity options like WiFi, Bluetooth, and 5G [308], [309]. These technologies are essential for enabling inter-device communication and advancing industries towards smarter operations. Achieving optimal performance and scalability requires substantial investment and integration of various technologies. Additionally, the increasing number of industrial devices generates large volumes of data, demanding scalable, cost-effective data processing and storage solutions. Efficient data handling increasingly relies on fog/edge and cloud computing technologies, as explored in Section VI-VI-C. Security systems play a critical role in mitigating vulnerabilities within interconnected Industrial AIoT systems that require persistent vigilance and substantial resources to ensure data integrity. Furthermore, power and energy systems require innovative solutions to maintain operational continuity alongside eco-friendly practices. The rapid technological evolution burdens maintenance and support services, necessitating ongoing staff training and proactive maintenance strategies to keep systems functional and up-to-date. Adhering to evolving industry standards and regulations is crucial for ensuring to manage the infrastructural needs to leverage AIoT potential within industrial settings.

Technological limitations significantly challenge the adoption and effective deployment of Industrial AIoT systems. Issues such as data privacy, security, and data integration significantly impact system vulnerability to breaches and unauthorized access. The lack of interoperability and standardization across IoT devices and platforms complicates data exchange, while constraints on network capacity and data processing limit scalability and sustainability. Additionally, integrating AIoT solutions with existing industrial systems presents compatibility and synchronization challenges. The reliability and availability of Industrial AIoT systems are at risk due to potential hardware failures, software bugs, and network downtimes, crucial for maintaining continuous operations. The high energy consumption of devices and the need for high-performance computing present additional challenges. Furthermore, particularly for AI techniques, the quality of results also highly depends on the quality of data available, data selection and its preprocessing as described in Sections V-V-A, V-V-B, and V-V-C, emphasizing the importance of robust data management practices. In addition, we also need a workforce with experience in IoT, AI, analytics, cybersecurity, and network administration for the effective deployment and maintenance of industrial AIoT systems. With new technologies emerging every day, the workforce has a big skill gap. Unlocking the revolutionary potential of Industrial AIoT requires a coordinated effort to address these limitations, which include improving the energy economy, developing more rigorous security measures, ensuring dependable real-time processing, and enhancing system interoperability.

H. AI TECHNIQUE SELECTION

In Industrial AIoT, the selection of an AI technique is crucial and requires a multifaceted approach. Various AI techniques are discussed in Section III. This selection is not just about identifying a technique that is technically capable of solving a given problem but also involves considering a suite of additional factors to ensure the chosen solution aligns with broader operational, ethical, and practical requirements. The relevance and suitability of an AI technique form the foundation of this selection process. For example, techniques such as kNN, SVM, and NB are well-suited for anomaly detection, which is a common challenge in industrial settings. Similarly, Transformer models and their variants are more suitable for processing sequential data and making long-term predictions based on time-series data [310]. FL is more suitable in a distributed industrial environment. Conversely, GANs and CNNs are preferred for tasks involving data generation and complex image analysis, respectively, showcasing the diverse applicability of AI techniques across various industrial domains [311]. Therefore, selecting a suitable AI technique according to the nature of the industrial problem is very critical. In the case of existing systems, the integration of these techniques into existing systems introduces another layer of complexity due to the thorough evaluation of integration, implementation, and infrastructural demands. It includes assessing the technical challenges and the resource implications of deploying these models, highlighting the importance of choosing AI solutions that are not only effective but also feasible within the given infrastructural constraints. Therefore, while selecting an AI technique, it is important and challenging to choose a technique that is more relevant or suitable and compatible with the existing system, in case there is any.

Another point to consider during the selection of AI techniques for an Industrial AIoT system is training complexity. DL techniques are computationally complex and require more training time, making them more difficult to train. Training complexity presents a significant challenge in Industrial AIoT due to the extensive computational resources and data required to develop effective models. Therefore, it is challenging to maintain a balance between computational cost and efficiency while selecting an AI technique [312]. The complexity of industrial processes and the vast amount of data generated necessitate sophisticated algorithms that are both resource-intensive and time-consuming to train. Moreover, the variable and dynamic nature of industrial environments complicates the training process since models must be adaptable and resilient to changing operating conditions. Once the appropriate AI technique is selected, the next challenge is optimizing the hyperparameters. Hyper-parameter selection is not only important but also a challenging task. The accuracy of the chosen AI technique is significantly influenced by the optimality of the selected hyperparameters. Various techniques for hyper-parameter selection, such as GridSearch CV, should be explored for better and more accurate results.

When selecting an AI technique, it is essential to also consider safety standards to minimize risks and unintended consequences of AI deployment in industrial settings. This includes ensuring robustness against adversarial attacks and resilience to errors. AI systems that operate in close collaboration with humans, through robots or other machinery, must adhere to both current and forthcoming safety standards. They need to be capable of handling inaccurate data robustly and offering low-latency responses to unexpected events, especially in managing safety-critical processes [313]. Moreover, the ethical implications of using AI techniques add more complexity to AI technique selection. During the deployment of Industrial AIoT systems and particularly AI technique selection, we should also consider the alignment of technological applications with ethical considerations and societal values. Ethical consideration involves a thorough examination of how AI technologies impact privacy, fairness, autonomy, and accountability within the industrial context. This aspect is important as the deployment of Industrial AIoT systems can significantly influence operational efficiency, environmental sustainability, and worker safety. Furthermore, in ethical AI, fairness and bias are the ethical considerations that must be addressed to achieve fair outcomes for all individuals [314]. Fairness can be described as the absence of bias or favoritism toward an individual or group based on their inherent or acquired characteristics [315]. This makes ethical oversight a challenging yet essential aspect of AI technique selection in industrial applications. AI algorithms are prone to biases in the data on which they are trained, resulting in biased outcomes in decision-making processes [314]. The AI techniques must prioritize transparency, allowing stakeholders to understand and evaluate the decision-making processes of Industrial AIoT systems. Moreover, continuous monitoring and adjustment of AI algorithms are necessary to adapt to changes and minimize adverse outcomes, ensuring AI technologies contribute positively to society while aligning with ethical standards and promoting equitable, safe, and sustainable industrial practices. Furthermore, selecting AI technique and further building and deploying Industrial AIoT solutions based on that technique also raises some societal challenges as well. It necessitates a thorough exploration of its potential societal impacts. This involves considering how these technologies may affect job displacement, economic disparities, and community well-being. While Industrial AIoT systems offer opportunities for efficiency gains and innovation, they also raise concerns about job displacement and inequality. The potential societal and further economical impact of Industrial AIoT based solutions are further explained in detail in Section VII.

By integrating all these considerations for selecting an AI technique, ranging from technical feasibility and operational efficiency to ethical responsibility, the process of selecting AI techniques for Industrial AIoT applications becomes a comprehensive evaluation, ensuring that the chosen methods are not only technically suitable but also aligned with broader goals of safety, transparency, and ethical integrity. Choosing

the wrong Industrial AIoT technique can lead to a range of issues, including unexpected results and the wastage of both time and financial resources. Therefore, selecting the most suitable Industrial AIoT technique for a specific issue is quite challenging.

I. INTERPRETABLE, EXPLAINABLE, AND TRUSTWORTHY INDUSTRIAL AIOT TECHNIQUES

The rapid adoption of AI techniques in industrial sector has revolutionized the industrial processes ranging from PdM to real-time monitoring and decision-making. However, the black box nature of many AI algorithms or techniques poses challenges in interpretability, explainability, and trustworthiness. Interpretability refers to the ease with which humans can understand the decision-making process of AI models [316]. In industrial settings, interpretable models are crucial for diagnosing issues, improving system performance, and ensuring safety. Techniques like LIME (Local Interpretable Model-agnostic Explanations) [317] and SHAP (Shapley Additive Explanations) [318] have been adapted to make complex models more interpretable without sacrificing performance. However, for the complex Industrial AIoT techniques or models, it is hard to make the interpretation and incorporate the interpretability.

Building on the concept of interpretability, explainability aims to provide human-understandable reasons behind AI decisions [319]. Explainability signifies the contribution of input features in overall model predictions. This is particularly important in critical industrial applications where understanding the rationale behind decisions can have significant implications. XAI techniques are being developed to make the internal workings of AI models transparent. However, the field currently lacks standardized metrics for explainability, making it difficult to evaluate and compare the effectiveness of different models objectively.

Beyond interpretability and explainability, the concept of trustworthiness in AI systems also encompasses aspects like reliability, robustness, and ethical considerations. Trustworthiness ensures the users confidence in a particular AI technique and its decisions. Explainability contributes to trustworthiness. The more explainable a model is, the greater the trust users will have in it. Techniques such as adversarial training and robust optimization are being developed to build trust in AIoT systems. Yet, the black-box nature of these models raises additional challenges, including ethical and regulatory concerns, especially when deployed in safety-critical environments.

Several case studies have demonstrated the effectiveness of interpretability and explainability techniques in real-world industrial applications. For example, interpretable models have been successfully deployed in PdM tasks [320], while explainable models have been utilized in quality control in manufacturing. Despite these advancements, the challenges posed by the black-box nature of many AI models make the incorporation of interpretability, explainability, and

trustworthiness not just a theoretical exercise but a practical necessity for the responsible advancement of technology in industrial settings.

In summary, the integration of interpretable, and explainable techniques that enhance trustworthiness in the industrial sector are crucial yet challenging due to the black-box nature of many AI models. Techniques like LIME and SHAP are advancing interpretability, while the development of XAI techniques aids in explainability. However, issues like the lack of standardized metrics for explainability and ethical concerns in safety-critical environments remain. Case studies demonstrate the potential of these techniques in applications like PdM and quality control, underscoring their practical necessity in responsible industrial AI development.

VI. FUTURE DIRECTIONS FOR INDUSTRIAL AIOT

The modern industry will benefit from Industrial AIoT as it improves efficiency and reduces costs. In this section, we present several important research directions that need to be thoroughly explored before Industrial AIoT can realize its full potential.

A. SENSOR FAULT DETECTION IN INDUSTRIAL AIOT

AI techniques have the potential to revolutionize decision-making processes in the industrial environment, addressing challenges related to PdM, resource management, and alert generation. This, in turn, can enhance overall safety and minimize financial losses. Industrial AIoT applications encompass a vast number of sensors, industrial equipment, and actuators. Sensors are responsible for gathering data, which is then processed for various purposes, including PdM-related decisions and effective human resource management [321], [322]. Research should focus on improving decision-making through intelligent analysis, particularly with sensor data. The reliability of sensors plays a crucial role in ensuring accurate, data-driven decisions within industrial settings. Factors such as aging, wear, improper calibration, and environmental conditions significantly contribute to sensor-related errors [323]. These environmental conditions include humidity, temperature fluctuations, weather changes, and atmospheric pressure. All these factors can greatly increase the likelihood of errors in sensor readings, malfunctions, or anomalies in sensor behavior. Faulty sensor values can potentially have a detrimental impact on decision-making, altering results and leading to incorrect conclusions. This underscores the critical importance of maintaining sensor accuracy and reliability. Sensor faults include drift, gain, noise, erratic, hard-over, stuck, spikes, and other random faults [324], [325], [326]. All these types of faults exhibit distinct patterns that are critical for effective detection and analysis. Bias in sensor readings can lead to misinterpretations, while gradual changes in measurements over time can compromise data accuracy. Noise in sensor readings can mask genuine trends, and extreme conditions can result in incorrect data. Furthermore, sensors that react to multiple environmental factors can complicate data interpretation.

Thus, detecting and mitigating sensor-related issues are essential steps in ensuring reliable data-driven decisions and formulating effective operational strategies.

In a multi-sensor industrial environment, each sensor can potentially develop any type of fault. Therefore, there is a pressing need to focus future research on multi-sensor, multi-class fault detection and identification, utilizing the most accurate and advanced techniques available such as DNN, CNN, and Transformer. Integrating sensor fault detection and identification with decision-making systems is crucial for enhancing system robustness and accuracy. This integration allows for real-time monitoring and immediate corrective action in response to sensor faults, thereby minimizing the impact of inaccurate data on critical decisions. Employing these techniques not only improves the reliability of the systems but also contributes to safer and more efficient industrial operations.

In scenarios where authentic industrial faulty sensor data is scarce or missing, a viable strategy involves injecting faults into healthy data for training and evaluating AI models. This technique presents two principal benefits [325]. Firstly, the artificial injection of faults into sensor measurements creates a well-defined “ground truth,” crucial for evaluating the performance and effectiveness of fault detection and identification techniques. Secondly, this approach offers the flexibility to modify both the intensity and frequency of faults in diverse scenarios. This adaptability is key, allowing for a comprehensive assessment and comparison of various methods under different fault intensity conditions. Furthermore, once faults are injected, generative AI models such as GANs, VAEs, and diffusion models can be employed to simulate and generate diverse fault scenarios. These generative models play a crucial role in enhancing the robustness of training and evaluation processes, as mentioned in Section VI-VI-G.

B. GREEN AND SUSTAINABLE INDUSTRIAL AIOT

Energy efficiency has emerged as a central research topic, especially in modern industrial infrastructures where heterogeneous machines consume large amounts of energy. These infrastructures incorporate sensors, devices, manufacturing machines, robots, and various industrial equipment. All these devices are connected together and enable seamless Machine-to-Machine (M2M) communication. Despite their innovative designs, these devices possess limited resources that can impact their lifespan. They continuously engage in sensing, computing, and communicating. While these activities are vital, they demand significant energy, increasing the environmental carbon footprint [327]. The integration of AI techniques into this context presents promising opportunities for more efficient industrial systems. Despite their effectiveness, these techniques are resource-intensive. They rely on large, powerful servers that consume substantial amounts of energy, highlighting the need to address the energy efficiency and sustainability aspects of AIoT solutions [304]. The impact

of energy efficiency, particularly in IoT-enabled industrial systems, is twofold: it affects device longevity due to resource constraints and increases the system’s carbon footprint through extensive sensing, computing, and communicating activities [327]. The extensive sensing and use of these devices lead to faster degradation and shorten the lifespan of these devices. From the device perspective, the devices are constrained in communication and computation, and energy consumption can shorten their life. On the other hand, from the whole industrial systems perspective, the industrial devices lead to excessive carbon footprints. It is crucial that Industrial AIoT solutions are energy-efficient and sustainable to mitigate negative environmental impacts.

Addressing the challenges related to energy consumption paves the way for future research focused on energy conservation in Industrial AIoT. This exploration must consider the environmental impact of deploying such technologies, underscoring the importance of developing lightweight AI models that are less energy-intensive, as detailed further in Section VI-VI-D. Edge computing, which processes data closer to its source, offers a promising approach to energy savings. It effectively reduces the need for data to travel long distances to centralized servers and conserve energy.

The drive to minimize the impact of computing technologies on the environment has led to the rise of environmentally sustainable computing, also known as sustainable IT or green IT. This approach seeks to develop eco-friendly solutions [328]. Sustainability in this context is intertwined with energy efficiency, where practices aim to reduce environmental impact by considering both the short and long-term effects on the environment [329]. It is closely linked with energy efficiency, featuring initiatives such as the use of eco-friendly servers and data centers, adaptive algorithms for optimizing device operations, and advancements in battery technology. These measures not only conserve energy but also extend the lifespan of industrial devices, promoting sustainability in industrial operations by reducing resource consumption and minimizing the carbon footprint.

At a macro level, future research should focus on comprehensive strategies to manage energy, harmonizing the requirements of machines, communication protocols, AI, and sustainable computing to ensure a sustainable Industrial AIoT ecosystem. This approach not only aims to reduce the carbon footprint of industrial operations but also ensures that the deployment of Industrial AIoT solutions contributes positively to environmental conservation efforts. The emphasis on sustainability, a core component of Sustainable Computing, addresses the long-term environmental impact by promoting the use of renewable energy sources and eco-friendly practices. Thus, Industrial AIoT is positioned to lead towards a more sustainable and greener future. By integrating energy efficiency, sustainability, and Sustainable Computing principles thoughtfully, the field of Industrial AIoT can advance towards an environmentally responsible and sustainable future, epitomizing the ideals of green industrial practices.

C. FOG/EDGE COMPUTING FOR INDUSTRIAL AIOT

Industrial processes involve heterogeneous sensors and devices that produce a large volume of data. Traditionally, these heterogeneous distributed devices in the industrial environment generate data and transmit it to the cloud server for processing. However, with the tremendous growth of industrial infrastructure, cloud computing does not always guarantee real-time and efficient data processing [330]. For example, in smart parking, cameras collect images of the parking spots and transmit them to the cloud server for processing to determine whether a space is vacant. In such cases, high bandwidth is consumed, increasing the overall processing delay [184].

Low latency, reduced bandwidth consumption, and higher system reliability are essential requirements in industrial systems [331]. Therefore, a fast and efficient mechanism is needed to process the data intelligently with low latency, reduced bandwidth, and high reliability. Edge computing, a relatively new computing paradigm, provides storage and computational resources at the edge of the network for analyzing and processing the data [332]. Edge computing uses devices with sufficient storage and processing capacity to intelligently process local data and then transmit it to the cloud server for longer-term usage. Essentially, edge computing improves the overall performance of the system by reducing operation costs and enhancing data security and privacy. Recently, cloud/edge computing architectures for industrial infrastructure have provided effective, efficient, and intelligent computing for industrial applications. In Industrial AIoT, edge computing also allows the deployment of AI models on edge devices, processing data with lower latency [190].

Fog computing is another concept similar to edge computing. It is a distributed computing paradigm that provides cloud-like services at the edge of the network [333]. The primary goals of fog and edge computing are to offer computational resources near the devices, reducing overall end-to-end delay, energy consumption, and network bandwidth usage. However, according to the National Institute of Standards and Technology [334], the fog computing paradigm is a horizontal, physical or virtual resource layer that exists between smart end-devices and traditional cloud computing environments. On the other hand, edge computing focuses on bringing computation and storage resources closer to data sources. Furthermore, fog computing introduces a new fog layer between the cloud server and edge devices. In recent studies, fog computing has been found beneficial for realizing the full potential of Industrial AIoT applications [15]. One pivotal issue associated with resource sharing in this context is load balancing. Achieving effective load balancing, where computational resources are evenly distributed, remains a significant challenge. While substantial work has been done in the past on load balancing for fog and edge computing [335], [336], [337], there is potential for further optimization using AI techniques. Addressing this issue

is essential to ensure that the Industrial AIoT ecosystem operates efficiently and without delays. Solutions to this problem would not only enhance system responsiveness but also improve the reliability of the devices and servers involved by preventing potential overloads.

The storage and processing capabilities of industrial devices are minimal. Some computation-intensive tasks require significant computational resources to carry out the task efficiently. With the integration of AI techniques, tasks can be intelligently offloaded to the edge or fog servers [194], [338], [339]. By leveraging AI, the decision-making process for where and when to offload can be optimized, ensuring that tasks are processed in the most efficient and timely manner. The task could be partially or completely offloaded based on the AI-driven decision. AI models can be trained to make these offloading decisions more efficiently, taking into account various parameters like device capabilities, network conditions, and task requirements. In the past, a significant amount of work has been done for task offloading using AI techniques such as DRL. In the future, AI techniques can be integrated with green Industrial AIoT to make intelligent and energy-efficient offloading decisions, further enhancing the efficiency of the entire system.

Some existing surveys describe the use of edge and fog computing in Industrial AIoT applications. For instance, Qiu et al. [331] presented the architecture, opportunities, and challenges of edge computing. Georgakopoulos et al. [340] described the role of IoT and edge computing in the manufacturing industry. Steiner et al. [341] introduced fog computing as a layer in the IIoT architecture. Azam et al. [15] proposed the fog-based architecture for Industry 4.0 and discussed the role of fog computing. Moreover, the authors also discussed the potential use cases and challenges of fog computing in IIoT. However, the applications of Fog/Edge computing for Industrial AIoT have not been sufficiently discussed in-depth, and it remains an open research issue.

In a nutshell, the evolving landscape of Industrial AIoT presents both remarkable opportunities and intricate challenges. As the integration of heterogeneous sensors and devices continues to grow, ensuring efficient data processing and resource sharing becomes paramount. The issues of load balancing and real-time data processing underscore the need for innovative solutions that can adapt to the dynamic nature of industrial environments. While paradigms like edge and fog computing offer promising avenues, there is a clear necessity for further research to refine these approaches and address the inherent challenges they present. The future of Industrial AIoT relies on the development of robust, scalable, and efficient computational frameworks. It is imperative for academia and industry to collaborate, focusing on devising advanced strategies and technologies that can truly harness the potential of AI-driven industrial automation. This would not only optimize system performance but also pave the way for a more sustainable, efficient, and intelligent industrial future.

D. LIGHTWEIGHT AI TECHNIQUES FOR INDUSTRIAL AIOT

The rapid evolution of the Industrial AIoT has resulted in a new era of smart manufacturing and automation. AI techniques are at the top, empowering devices to make intelligent decisions, optimize operations, and enhance overall system performance. However, the inherent complexity of traditional, computationally rich, powerful AI models often demands high computational resources. This makes them unsuitable for the resource-constrained environments typical of industrial devices. Consequently, there is a growing interest in lightweight AI techniques tailored for the Industrial AIoT.

The need for Lightweight AI is underscored by the characteristics of industrial devices. These devices, ranging from sensors to robots, typically possess limited computational capabilities and energy reserves. Deploying complex AI models on them can quickly exhaust their batteries and shorten their operational lifespan. Moreover, continuously transmitting vast amounts of data to a central server for processing can introduce latency. This delay in decision-making can be detrimental, especially in time-sensitive industrial operations where real-time responses are crucial.

Despite the use of edge and fog computing in resource-constrained environments, challenges remain. To address these challenges, lightweight AI models have been designed to be both compact and efficient. They are optimized to require less memory and computational power, making them particularly suited for edge devices within the Industrial AIoT environment. By enabling local data processing, these models significantly reduce data transmission needs, ensuring faster response times and conserving energy [342]. This local processing not only enhances system efficiency but also ensures real-time decision-making, a pivotal aspect for many industrial applications.

Several strategies have been employed to achieve this lightweight nature, with model compression emerging as a particularly promising technique [343]. Model compression seeks to reduce the size of AI models without significantly affecting their performance [344]. One of the primary techniques in this domain is pruning, which involves eliminating certain parts of the neural network, such as weights or neurons, that contribute minimally to the output of the model. This streamlining process retains the most critical components of the network, ensuring efficiency without a substantial loss in accuracy. Another technique, quantization, reduces the precision of the numbers used within the model. Models might use 8-bit integers instead of 32-bit or 64-bit floating-point numbers. The result is smaller model sizes and faster computation, with little impact on performance. Additionally, there is knowledge distillation, a process where a smaller model, often referred to as the student, is trained to replicate the behavior of a larger, more complex model, known as the teacher. The student model learns from the outputs of the teacher model rather than the original data, enabling it to achieve comparable performance with a fraction of the size. Given the potential of these techniques, future

research could delve deeper into their application in the industrial context. There is an opportunity to craft lightweight AI models tailored for specific industrial domains, ensuring that these models are finely tuned to address the unique challenges and data types inherent to each sector.

Moreover, the emergence of Neural Architecture Search (NAS) offers another dimension of optimization [345]. NAS involves an automated search for the most suitable neural network architecture for a specific task, ensuring that the resulting models are both efficient and effective. Coupled with this is the emphasis on hardware-aware AI, which highlights the importance of developing AI models with a clear understanding of the specific hardware they will be deployed on. This convergence of AI research and hardware design can lead to models and devices that operate in perfect harmony, maximizing efficiency.

All things considered, the expanding horizon of the Industrial AIoT demonstrates the importance of lightweight AI techniques. By focusing on the development and deployment of these efficient models, the industry can ensure longer device lifespans, rapid decision-making, and sustainable energy consumption. As we move forward, the synergy between lightweight AI and the Industrial AIoT will undoubtedly pave the way for a more efficient and sustainable Industrial AIoT future.

E. FEDERATED LEARNING: A PARADIGM SHIFT FOR INDUSTRIAL AIOT

AI techniques have led to in a new era of intelligent industries, offering groundbreaking capabilities in data analysis, prediction, and automation. However, the traditional AI models relies on centralized data repositories often limits its potential, especially in scenarios where data is limited, sensitive, or distributed across multiple locations. Recognizing this limitation, Google introduced FL as a novel approach to decentralized ML [346].

Unlike conventional AI methodologies where data is centralized for model training, FL reverses this paradigm. In FL, data remains on decentralized devices, such as mobile phones or industrial sensors. These devices, often referred to as nodes, train local models on their data without sharing the raw data itself. Instead, they periodically share model updates or parameters with a central server, which aggregates these updates to improve a global model. This approach not only preserves data privacy but also leverages the collective intelligence of all participating nodes.

The integration of FL into industrial applications offers numerous potential benefits. Foremost among these is the enhancement of data security and privacy. By keeping data localized, FL minimizes the risks associated with data breaches or unauthorized access [105]. Additionally, by reducing the need for continuous data transmission to a central server, FL can significantly reduce communication costs, lower latency, and decrease power consumption which are all critical considerations in an industrial context.

In the field of smart industries, FL's decentralized approach facilitates real-time, intelligent decision-making. Local models can quickly process data and make decisions, while periodic synchronization with the global model ensures that these decisions benefit from broader insights and trends. Moreover, the synergy between FL and other emerging technologies, such as edge computing in the industrial environment, offers promising solutions to real-time challenges in data processing and decision-making. Given the inherently distributed nature of industrial environments, FL emerges as an especially fitting tool, enabling seamless collaboration and data-driven insights without centralizing sensitive or proprietary data.

The applications of FL in the Industrial AIoT landscape are vast. From smart automotive systems that leverage FL for real-time traffic analysis and prediction, to smart healthcare applications where patient data privacy is paramount, and onto smart manufacturing setups where production data from multiple facilities can be used to optimize processes without compromising proprietary information.

However, the journey towards widespread adoption of FL in industrial settings is not without challenges. Among these are data heterogeneity, communication overhead, model aggregation complexity, client node selection, resource management, and etc. Addressing these challenges requires a multi-faceted approach, combining advances in ML, network communication, and industrial engineering.

As the field of FL continues to evolve, several research directions emerge as pivotal. The development of efficient algorithms for client node selection, ensuring fairness and representativeness, is crucial. Additionally, strategies for efficient resource management, both in terms of data and computation, need to be refined. As industries increasingly adopt IoT, the integration of FL with edge computing and real-time analytics will be a focal point of research. Furthermore, the development of robust security protocols to safeguard the FL process, from local model training to global model aggregation, will be essential to ensure the integrity of the entire system. Addressing the challenge of model heterogeneity, ensuring that diverse local models can be effectively aggregated without loss of accuracy or performance, will also be a key area of future research.

In summary, while challenges persist, the potential of FL in revolutionizing the Industrial AIoT landscape is immense. By addressing the current limitations and focusing on the highlighted future directions, the research community can pave the way for a more decentralized, efficient, and secure industrial future.

F. TRANSFER LEARNING

Transfer learning has emerged as a powerful technique in the domain of AI, especially in scenarios characterized by limited data availability [347]. In traditional ML, models are trained from scratch, requiring vast amounts of labeled data to achieve satisfactory performance. However, in many Industrial AIoT applications, obtaining such extensive labeled

datasets can be challenging, time-consuming, and expensive as described in Section V. In this case, transfer learning is important. By leveraging knowledge from a pre-trained model, developed on a different but related task, transfer learning allows for the adaptation of this model to a new task with minimal data. This is particularly beneficial for the Industrial AIoT, where conditions and requirements can vary greatly across different sectors and even individual industrial setting.

Significant work has been done in the domain of transfer learning and industrial automation [221], [348], [349]. However, several future directions exist for transfer learning in the Industrial AIoT. One promising avenue is the exploration of cross-domain adaptation, where models trained in one industrial setting, such as automotive manufacturing, can be adapted for another, such as chemical processing or textile production. This could facilitate rapid AI deployment across varied industrial landscapes. Another direction is the development of more sophisticated domain adaptation techniques that can handle the particularities of different industrial environments, ensuring that transferred knowledge is both relevant and effective. Additionally, as the Industrial AIoT continues to grow and evolve, there will be a need for continuous transfer learning approaches that can dynamically adapt to changing conditions and operational requirements without the need for frequent retraining. Collaborative efforts between AI researchers, industrial practitioners, and domain experts will be essential in driving these innovations, ensuring that transfer learning remains at the forefront of the Industrial AIoT evolution.

G. GENERATIVE AI FOR INDUSTRIAL AIOT

Generative AI models, such as VAEs, Autoregressive models (ARs), GANs, and Diffusion models, can play a crucial role in Industrial AIoT applications. Generative AI models should be explored for various purposes in Industrial AIoT, including PdM applications, sensor synthetic data generation, anomaly detection, network traffic prediction, and more. ARs stand out for their ability to predict future values in a data sequence based on past values [283]. This predictive capability can be leveraged in Industrial AIoT for forecasting and anomaly detection, particularly in scenarios where historical data is crucial for predicting future trends or identifying irregular patterns. VAEs offer another avenue for innovation. These models, known for their proficiency in data generation and anomaly detection, can also be particularly useful in dealing with missing or incomplete data in industrial settings. VAEs can reconstruct missing or faulty sensor data, ensuring more accurate and complete datasets for AI model training and decision-making. The role of GANs also remains significant. Their capacity to generate synthetic data that closely mimics real-world scenarios is invaluable, particularly in addressing issues like class imbalance and limited labeled data. They can simulate a range of conditions and scenarios, enhancing the robustness of AI models in industrial applications.

Diffusion models are the emerging class of Generative AI models and have shown the tremendous performance in many applications, including image generation, video generation, and data generation [350]. A diffusion model consists of two key stages: the forward diffusion stage and the reverse diffusion stage [351]. In the forward diffusion stage, the model progressively alters the input data by introducing Gaussian noise in a series of steps. Conversely, in the reverse diffusion stage, the model's objective is to reconstruct the initial input data by systematically undoing the diffusion process, step by step. Diffusion models are highly regarded for their ability to produce high-quality and diverse samples.

Advanced AI techniques such as LSTM, CNN, and Transformer require a large amount of labeled data for training. As described in Section V-V-A, one of the most significant challenges in Industrial AIoT is the unavailability or insufficiency of data for training robust AI models, data imbalance, and data quality. In the case of limited available data or imbalanced data, data generation and augmentation are found to be an effective solution. Generative AI models work by learning the complex patterns in the data and generating new data that resembles the real data. GANs have gained immense popularity in recent years for generating images, videos, and various other types of data [352], [353], [354]. These models can be used to generate synthetic industrial data that closely resembles real-world scenarios, thereby supplementing existing datasets [283]. For instance, in the context of PdM, a CNN-based GAN approach could be used to simulate equipment conditions, producing data that mirrors real-world degradation patterns [173]. Generative AI models can learn the distribution of the data and generate similar data. For instance, in cases of class imbalance, GAN-based models have proven effective in addressing these problems [355], [356]. Similarly, the authors in [357] used the diffusion model to address the challenge of the rare class in network intrusion detection. By utilizing the diffusion model, the data was generated for the rare class, which effectively balanced the training dataset across different classes and improving classification technique performance.

In the industrial environments where limited labeled data is available and faulty states occur infrequently, generative AI models can be used to synthesize realistic data, thereby countering the natural imbalance where datasets are heavily skewed towards normal operational states. This capability is crucial for training robust PdM models that can accurately identify and anticipate equipment failures. For instance, generative models can be tailored to simulate specific sensor-related faults, such as bias, gain errors, and erratic sensor behavior. This allows for a more comprehensive understanding and testing of Industrial AIoT systems under various fault conditions, enhancing the system's resilience and reliability. By generating synthetic data that mimic real-world anomalies and faults, generative AI models play a pivotal role in advancing the predictive analytics and maintenance strategies in Industrial AIoT.

The challenge of missing sensor data, often due to transmission errors or hardware malfunctions, can also be tackled using generative AI models. Instead of traditional imputation methods, which might introduce biases, generative AI models such as GANs [180] and ARs can generate data that aligns with the underlying distribution of the original dataset, ensuring more accurate and consistent imputations. Furthermore, there has been relatively limited exploration of diffusion models for addressing missing data imputation tasks. Zheng et al. [358] proposed a diffusion model based approach for the missing value imputation in tabular data. Hence, these types of approaches should be explored in Industrial AIoT applications for missing data imputation.

Generative AI models have become integral in industrial settings for applications like anomaly detection, trust-boundary protection, platform monitoring, and network traffic prediction. They can learn normal data distributions, key for identifying anomalies and verifying data conformity. In trust-boundary protection, GANs are used to create adversarial samples, enhancing security through rigorous testing of network segments organized by user privileges. For platform monitoring, especially in industrial settings, these models tackle challenges like fault data scarcity and imbalanced datasets, aiding in PdM. In network traffic prediction, their ability to improve traffic classification accuracy and manage wireless channel distributions is invaluable, even in scenarios with limited labeled data.

In the development of Generative AI models for Industrial AIoT applications, a range of challenges emerge, highlighting the need for continuous research and innovation. These models face limitations due to their fixed network architectures and stringent input requirements, making advanced preprocessing essential to manage the noise and abnormal data typical in Industrial AIoT. Generative AI models, especially in structured probabilistic models and unsupervised tasks like GANs, often suffer from weak discriminative capabilities, a common problem known as mode collapse [359]. This collapse results in the generation of a limited range of outputs, failing to capture the broader distribution of the training data and leading to repetitive and non-diverse outputs. Recent efforts have aimed to address the mode collapse problem [360], [361], but more research is needed in the future to solve this problem and enhance GAN efficiency. To tackle these challenges effectively, we should also explore the integration of multiple Generative AI model types, including diffusion models, semi-supervised approaches, and innovative solutions. Moreover, the interpretability of Generative AI models remains a challenge, crucial for understanding the semantic meaning of the generated data. The generalization ability of these models is also limited, often only generating data similar to their training sets. This calls for continual learning strategies to improve adaptability. Furthermore, the vulnerability of generated data to detection techniques and the privacy concerns raised by using large real-world datasets in Industrial AIoT necessitate robust

privacy-preserving mechanisms, possibly through distributed GANs and diffusion models. Lastly, the complexity and volume of Industrial AIoT data demand the development of more efficient, lightweight Generative AI model architectures capable of handling multi-source data and optimizing energy consumption, especially in decentralized environments.

Furthermore, it is essential to increase the exploration of generative AI models in various facets of Industrial AIoT, including PdM, synthetic data generation, anomaly detection, addressing data imbalance issues, and generating faulty data in the context of industrial AIoT. These models have the potential to revolutionize these areas and contribute significantly to the advancement of Industrial AIoT applications. Addressing these challenges, including ethical concerns in generating realistic yet fake content, will be crucial for advancing Generative AI models, including diffusion models, for more effective and efficient Industrial AIoT applications.

In conclusion, the future of Industrial AIoT is significantly shaped by the integration of generative AI models like VAEs, ARs, GANs, and Diffusion models. These models are instrumental in addressing data challenges such as imbalances and anomaly detection, and in generating synthetic data for improved training and predictive accuracy. Their effective use is crucial for realizing the full potential of AI in industrial automation, ensuring enhanced efficiency and reliability in operations.

H. EXPLAINABLE AI FOR INDUSTRIAL AIOT

The integration of AI into the industrial sector emphasizes the use of models that are interpretable, explainable, and trustworthy. As the field of Industrial AIoT evolves, several pivotal research directions have emerged [362]. Advanced interpretability techniques are essential, especially as AI models, particularly DL architectures, grow in complexity. One significant challenge that arises is when predictions are made after imputing missing data. Current XAI techniques often struggle to recognize biases introduced by imputation, which can lead to misleading or inaccurate explanations [291]. Addressing this limitation is crucial to ensure that the explanations provided are both accurate and reflective of the underlying data processes.

The development of unified frameworks that integrate various explainability techniques can offer a comprehensive perspective on model decisions, providing to diverse industrial requirements [363]. Trustworthiness verification methods, which assess the reliability, robustness, and ethical considerations of AI models, will be paramount in ensuring the responsible deployment of AI in critical industrial settings. Furthermore, developing effective human-AI collaboration is crucial. This involves developing mechanisms for AI systems to transparently communicate their decision-making processes and for human operators to provide iterative feedback. Ethical and regulatory frameworks, developed in collaboration with ethicists, industry experts, and AI researchers, will guide ethical deployment of Industrial

AIoT techniques in safety-critical applications. Emphasizing domain-specific interpretability can address the specific complexities of different industrial sectors, such as energy or transportation. Moreover, as real-time decision-making becomes more prevalent, techniques that offer instant AI explainability without compromising decision speed will be invaluable. Lastly, integrating feedback loops, where AI models continuously learn from both their environment and human expertise, can further support trustworthiness, ensuring models remain updated and in sync with industrial dynamics. In summary, the future of Industrial AIoT depends on a balanced combination of efficiency, transparency, reliability, and ethical considerations, guiding the field towards responsible and impactful advancements.

VII. SOCIETAL AND ECONOMIC IMPACT OF INDUSTRIAL AIOT

The Industrial AIoT techniques are revolutionizing the industrial sector. By summarizing and analyzing Industrial AIoT techniques presented in Section III, we draw our thoughts and in-depth analysis about societal and economic impact of Industrial AIoT techniques presented in the subsequent sections.

A. INDUSTRIAL AIOT AND THE WORKFORCE: JOB DISPLACEMENT, SKILL REQUIREMENTS, AND TRANSFORMATION

The use of AI technologies in various industries is increasing to bring automation and efficiency to industrial processes. As the use of Industrial AIoT techniques continues to rise, they aim to bring more automation to industries and replace human workers [364]. Significant concerns have been raised about potential job replacements and the negative impact on job opportunities. Throughout history, technological developments have always resulted in job displacement while also offering new opportunities. As stated in [364], the McKinsey Global Institute forecasts that automation could displace up to 800 million jobs by 2030, necessitating significant retraining for 375 million people, while the World Economic Forum anticipates that AI will create 133 million new jobs but displace 75 million by 2022.

The increasing use of Industrial AIoT across various industries, like transportation, retail, finance, manufacturing, and in digital platforms such as autonomous driving, online advertising, and social media management, has significantly automated and transformed these sectors. This automation profoundly impacts job landscapes. In transportation, AI enhances route optimization and safety but shifts job demand from drivers to AI system maintenance and data analysis [365]. Retail benefits from personalized shopping and inventory management, demanding more data scientists than traditional sales roles. Finance sees AI revolutionizing fraud detection and customer service, reducing roles in traditional banking but increasing demand for AI expertise. Manufacturing efficiency is boosted through PdM and robotics, evolving jobs from manual labor to technological

oversight. The evolution of Industrial AIoT techniques opens avenues for new job creation, particularly in areas requiring sophisticated analytical skills, AI model development, data interpretation, and system maintenance. With the increased use of LLMs and advancements in AI, tasks like content creation, coding, and software engineering are being automated, impacting jobs not only in the industrial sector but across all fields. As stated in [366], it is estimated that the adoption of LLMs will affect at least 10% of work tasks for around 80% of the U.S. workforce, and at least 50% of jobs for about 19% of workers. Similarly, in the industrial sector, the use of AI techniques, particularly the increased use of LLMs, can cause job displacement.

With the continuous use of AI in the industrial sector, there is a potential for job displacement among low-skilled workers. AI is also automating tasks that used to be performed by high-skilled workers. For low-skilled workers, automation of routine manufacturing tasks, like packaging and basic assembly operations traditionally done by workers on assembly lines, could be taken over by robotic systems. For high-skilled workers, tasks such as equipment condition monitoring, root cause analysis of defects, and predicting failures can be performed efficiently through AI algorithms. Another example for high-skilled workers is the automation of financial analysis and investment decisions, roles traditionally performed by financial analysts. AI algorithms have the capability to analyze vast amounts of market data, identify trends, and make investment recommendations, potentially outperforming human analysis in speed and accuracy. AI in the industrial sector can create new jobs, such as AI system designers and engineers who develop and refine AI technologies, data scientists who analyze and interpret complex datasets to improve decision-making, and AI engineers who ensure AI applications adhere to ethical guidelines. Additionally, maintenance specialists who oversee the operation and maintenance of AI systems to ensure they function correctly, as well as roles in AI training and support, where individuals teach AI systems to recognize patterns or provide technical support for AI product users, are also emerging. The shift in skill requirements is one of the most significant impacts of Industrial AIoT on the workforce. Modern industrial workers need not only traditional technical skills but also digital literacy, including proficiency in data analytics, knowledge of AI and IoT systems, and the ability to interact with advanced machine interfaces. To thrive in an AIoT-driven industrial environment, current and future workers need access to a robust educational and vocational training framework.

In summary, Industrial AIoT presents both challenges and opportunities. By preparing for the change in work requirements and the transformation of jobs ahead of the curve, we can maximize the benefits of Industrial AIoT. The shifts in job displacement, skill requirements, and job transformation inherently influence societal structures, urging a shift towards a knowledge-based economy and necessitating a societal adaptation to new forms of employment and education. It is

the goal of technology to advance humanity rather than hinder it in the future. This change demands a shift from traditional job structures and promises a future powered by a combination of machine power and human intelligence, thereby intertwining technological advancement with societal progress.

B. ANALYSIS OF MARKET TRENDS, ECONOMIC IMPACT, AND ROI OF IMPLEMENTING AIOT SOLUTIONS IN INDUSTRY

Industrial AIoT approaches are transforming industries by enabling smarter, more effective, and efficient operations. They bring technological advancement, transform economic landscapes, and provide a considerable return on investment (ROI). ROI measures the net income ratio to investment, demonstrating the efficiency and profitability of an investment by comparing the return to the project's cost.

Industrial AIoT solutions are experiencing significant expansion. According to a report [367], the AI-enabled IoT market size is predicted to expand from USD 5.1 billion to USD 16.2 billion by 2024, at a CAGR of 26%. This growth is driven by the demand for more efficient operations, greater safety measures, and personalized consumer experiences in areas such as manufacturing, healthcare, agriculture, and logistics. The massive amount of data generated by industrial devices requires extensive analysis to find meaningful insights, raising the requirement for AI. As a result, firms are increasing their expenditures on AI-powered goods to stay competitive and fulfill changing consumer expectations.

The economic impact of Industrial AIoT is multifaceted, increasing productivity and reducing operational costs. Through the automation of routine work tasks and optimization of operations, productivity can significantly increase while reducing operational expenses. For instance, AI-enabled PdM [310] can predict the lifetime of equipment and equipment failures before they occur, consequently minimizing downtime and repair costs. A study by Deloitte estimates that PdM increases equipment uptime by 10% to 20%, reduces overall maintenance costs by 5% to 10%, and cuts maintenance planning time by 20% to 50% [368]. Furthermore, the economic impact extends to job creation in sectors that design, implement, and maintain Industrial AIoT systems. This shift necessitates workforce retraining and upskilling, highlighting the importance of educational and vocational training programs tailored to the needs of an AIoT-driven economy.

The ROI for deploying Industrial AIoT solutions is significant. Industries that implement Industrial AIoT technology can expect considerable increases in operational efficiency, product quality, and customer satisfaction. These advantages directly result in cost reductions, increased income, and greater market competitiveness. Furthermore, the ability of Industrial AIoT solutions to improve decision-making and introduce new business models leads to new revenue streams for businesses. However, the ROI depends on successfully installing Industrial AIoT systems that require significant

investment in technology, training, and change management. Companies proactively investing in Industrial AIoT solutions and developing an innovative culture are more likely to realize their full potential and ROI.

Beyond economic considerations, the societal impact of Industrial AIoT is significant. It has improved quality of life through smart healthcare systems, enhanced safety measures in workplaces, and more sustainable environmental practices. For example, Industrial AIoT applications in smart cities can optimize traffic flow, reduce energy consumption, and improve public safety. However, the widespread adoption of Industrial AIoT solutions also raises ethical and privacy concerns that must be addressed through robust policy frameworks and responsible technology use.

In summary, industries stand to gain much from Industrial AIoT, which will also change social standards and yield significant economic returns. By understanding market trends, the economic impact, and the ROI of implementing AIoT solutions, businesses can effectively navigate the potential and challenges of this technological growth. The emphasis should be on leveraging the advantages of Industrial AIoT solutions while minimizing any potential drawbacks through ethical business conduct, worker development, and environmentally friendly operations.

C. REGULATORY AND POLICY FRAMEWORK FOR INDUSTRIAL AIOT

The Industrial AIoT solutions are revolutionizing various aspects of our lives, from smart transportation to healthcare and the food industry. While these solutions offer substantial benefits, they also pose challenges, as discussed in Section V. These challenges necessitate the establishment of a comprehensive regulatory and policy framework to ensure the effective development and deployment of Industrial AIoT systems. Striking a balance between fostering innovation and ensuring the ethical, secure, and compliant use of these technologies is crucial. Therefore, the implementation of rules, standards, regulations, and policy frameworks is essential for maximizing the benefits of these technologies while minimizing their challenges [369].

Data privacy, cybersecurity, and product safety are critical issues that should be addressed by the regulatory and policy framework. The General Data Protection Regulation (GDPR) laws, established by the European Union [370], should be applied to Industrial AIoT solutions to ensure data security and privacy. These laws are essential for managing and securing personal data. Similarly, in the healthcare industry, the Health Insurance Portability and Accountability Act (HIPAA), a US legislation, was created to protect patient data and ensure its security and privacy [371]. When developing and deploying healthcare-based Industrial AIoT solutions, HIPAA rules must be implemented to protect patient data privacy.

To establish a robust regulatory and policy framework for Industrial AIoT, several key principles must be considered. Transparency and explainability are crucial aspects.

Industrial AIoT system decision-making processes should be explainable, transparent, understandable, and trustworthy to users, as described in Section V-V-I. Consequently, designers and developers must be held accountable for any biases introduced into their algorithms. Bias mitigation is essential in systems that could potentially lead to discrimination. For example, when systems are used to screen applicants for manufacturing industry positions, decisions should be based on fair and unbiased data to ensure all candidates are evaluated equally. There is a risk that these systems could perpetuate discrimination if the AI algorithms are not well-designed. For instance, if a system uses historical recruiting data that reflects past discriminatory behaviors, it may become biased against particular demographic groups. Qualified candidates from underrepresented backgrounds may score lower than their peers due to biased data rather than their actual skills or potential. Addressing such biases through explainability, transparency, and bias mitigation techniques is necessary to prevent discrimination and ensure fairness in Industrial AIoT systems. This requires incorporating bias assessment and auditing processes into Industrial AIoT development practices. Ethical guidelines should also be established within regulatory frameworks, emphasizing principles like fairness, non-discrimination, and respect for human rights in Industrial AIoT systems development and deployment. Violations of these guidelines should carry penalties to uphold ethical standards.

To enhance the integrity of Industrial AIoT solutions, aligning with international standards is essential. Compliance with these standards facilitates interoperability, strengthens security measures, and ensures quality across different systems and regions. Recommendations from standardization organizations such as the International Organization for Standardization (ISO) and the Institute of Electrical and Electronics Engineers (IEEE) are vital. For example, ISO/IEC JTC 1/SC 41 focuses on standards for the IoT and related technologies, providing guidelines that ensure interoperability and security. Similarly, IEEE has numerous standards related to AI and IoT, such as IEEE P7000, which addresses ethical considerations in system design. Furthermore, worker safety policies must be rigorously defined and implemented to safeguard individuals in environments where Industrial AIoT systems are deployed. Such policies should encompass comprehensive safety protocols and training programs, minimizing risks and creating a secure working environment.

In summary, developing a detailed regulatory and policy framework for Industrial AIoT solutions is imperative for balancing innovation with ethical, secure, and compliant technology use. This framework should address data privacy, cybersecurity, product safety, and incorporate international standards and worker safety policies, ensuring the responsible and sustainable advancement of Industrial AIoT technologies. Such a framework is vital for maximizing positive societal impacts, as it ensures that these technologies drive industrial efficiency and economic growth while enhancing

the quality of life and equity in society, paving the way for a future where technology serves all segments of society equally.

VIII. CONCLUSION

The intersection of IIoT and AIIoT has led to a new computing paradigm, namely, Industrial AIIoT, which has shown tremendous potential for making industrial processes more efficient. In this paper, we first review state-of-the-art ML and DL techniques for Industrial AIIoT. Specifically, the ML techniques that are covered include kNN, SVM, NB, DT, RF, ANN, and FL. The DL methods surveyed in this paper include AE, CNN, RNN, RBM, GAN, DRL, Transformer, and LLMs. Furthermore, we present the key applications of Industrial AIIoT, such as real-time manufacturing, smart healthcare, waste management, and intelligent transportation. In addition, we describe the challenges faced by Industrial AIIoT applications, including data availability, technique selection, and effective collaboration among heterogeneous devices. Finally, we highlight important future research directions in Industrial AIIoT, such as sensor fault detection, green Industrial AIIoT, and generative AI for Industrial AIIoT. We believe that this comprehensive survey will be helpful for Industrial AIIoT practitioners and researchers.

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